Spatial–Channel Selective State Space Models and Pruned Cross-Task Feature Transfer for Building Height Mapping From Aerial Images and DSMs

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***Abstract*—Building height mapping is widely coupled with footprint segmentation in a unified multitask learning (MTL) framework. In this study, we comprehensively explore the designs of the shared encoder, the bottleneck, and the decoders' cross-task feature sharing in MTL for precise building height mapping, and propose Mamba-based building height estimation network (MBHNet). First, we discovered that Mamba, a state space model (SSM) variant with selective feature extraction capacity, is particularly suited for inferring building heights. However, in previous remote sensing (RS) related studies, only the global feature correlation capacity, the same as Transformer, is utilized. We design a local-then-global feature encoder, locally sensitive SSM (LS-SSM), which integrates neighborhood attention (NATTEN), input discriminant enhancer (IDE), and spatial-dimension selective SSM, ensuring rich local-to-global information extraction and strong adaptability for MTL tasks. Second, a channel-dimension selective SSM (CSSM)-based bottleneck is introduced to model interchannel dependencies, in which, unlike previous works, the channel-wise information is not compressed and fully preserved. Third, considering the redundancy of the commonly used full feature sharing strategy across different tasks, we develop a K-top relevant feature mining (KTM) strategy in the decoding stage to enable selective and pruned cross-task feature transfer. Extensive experiments on three benchmark datasets demonstrate that MBHNet consistently outperforms recent methods, achieving very significant relative gains of 12.7%, 32.7%/40.2%, and 32.8% in mean absolute error (MAE) for building height estimation on the GeoNRW dataset, Tainjin Test Set-I/II, and the ISPRS Potsdam dataset, respectively. The datasets used in this study will be made publicly available at: https://github.com/yudawen/MBHNet**

***Index Terms*—Building height estimation, cross-task feature transfer, digital surface model (DSM), multitask learning (MTL), state space model (SSM).**

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I. Introduction

BUILDING height data reflects the status of urban development and offers essential insights into urban processes [1], [2]. Building height information also supports a wide range of analyses and applications, including 3-D building reconstruction [3], urban volume ratio analysis [4], population distribution mapping [5], and climate studies [6]. However, accurate building height data remains limited nowadays, especially at high spatial resolutions. Traditional field surveys are both labor-intensive and time-consuming, and are unable to deliver high-frequency updates or large-scale coverage [7], [8]. With the advancement of Earth observation technologies, high-resolution remote sensing (RS) data, characterized by extensive spatial coverage, has become a cost-effective alternative for building height mapping [2]. Over the past two decades, numerous automatic height estimation methods have been developed based on optical imagery [9], synthetic aperture radar (SAR) [10], and LiDAR [11], and digital surface models (DSMs) derived from optical imagery or LiDAR [12], [13].

Driven by the powerful feature extraction capabilities of deep learning (DL), the direct estimation of building heights from monocular or multiview images using prevalent multitask learning (MTL) architecture has attracted considerable attention in recent years [10], [14], [15]. In these works, height prediction is typically coupled with building footprint segmentation to ensure consistency between height discontinuities and building contours. Monocular height estimation is inherently an ill-posed problem, relying solely on visual cues such as semantic classes, object scales, and spatial contexts to recover height information, which often leads to low-confidence and coarse-grained predictions. DSM provides complementary and necessary vertical information for optical images. Their combination (e.g., through channel concatenation) has been proven effective by mitigating the limitations of DSM-only or image-only methods [16], [17]. Therefore, in this study, we explore the advanced MTL model based on the combined DSM and image inputs, with greater emphasis on the architecture and components of the model.

As illustrated in Fig. 1, building height estimation using MTL architectures has evolved from early simple versions [14] to increasingly sophisticated designs [9], driven by ongoing research in the RS community. Replacing a single decoder with dual-branch decoders and introducing cross-task feature interaction has demonstrated effectiveness in improving build-

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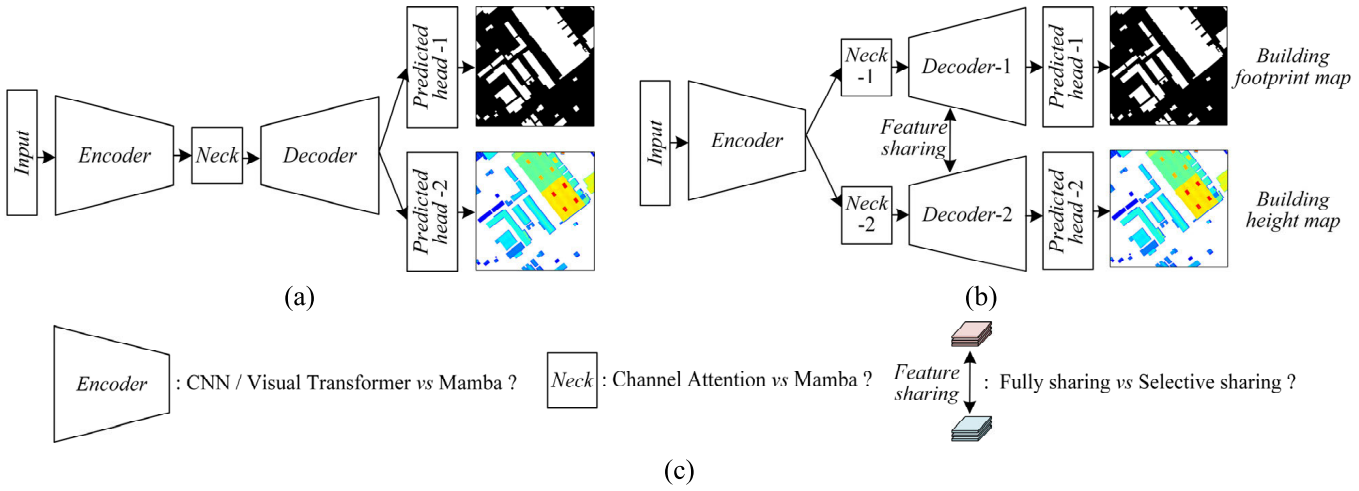


Fig. 1. Evolution of building height mapping pipelines from the early version (a) to increasingly sophisticated variants (b). (c) Currently underexplored parts: including the limited exploration of the recently proposed Mamba for feature encoding compared to CNNs and visual transformers, the untapped potential of Mamba for feature decoupling as an alternative to channel attention, and the architectural design of cross-task feature sharing (full or selective).

ing height mapping performance. More advanced CNN or visual transformer backbones also yield gains. However, significant challenges remain: First, the commonly adopted full feature sharing between dual branches may cause adverse feature recoupling during the decoding stage. Second, the recently popular Mamba [18] remains unexplored, despite its strong potential for feature encoding and decoding. In this work, we comprehensively investigate how to build an advanced MTL model for building height mapping, addressing both architectural design and component-level innovations.

Mamba as Encoder and Bottleneck: We observed that the Mamba [18], a new architecture evolved from the state space model (SSM) [19], is particularly suited for height inference due to its two key attributes, selective feature extraction and long-range feature aggregation. Building height estimation requires selective identification of relevant features, i.e., rooftop and ground, to compute accurate height values, while ignoring other adverse features like trees and cars, which aligns with Mamba's first attribute; long-range feature aggregation capacity can better recognize bare ground pixels, which are sometimes far away from a building footprint due to the interference from vegetation or other buildings. Moreover, visual Mamba (VMamba) [20], which creates multiple scanning routes within 2-D images, shows strong interpretability, resembling traditional scan-line-based terrain filtering methods [21] but without the limitations of empirical settings and computational inefficiency.

Despite the aforementioned advantages, directly integrating Mamba for building height mapping may yield suboptimal results. Mamba is characterized by an input-dependent selective scanning mechanism, that is, the model selectively remembers or ignores current inputs depending on their content. Previous RS-related studies [22], [23] only applied Mamba's global feature correlation capability but ignored this characteristic. We argue that, for RS images with complex internal variations of certain objects and rich spatial details, enhancing local feature discrimination should be prioritized before applying Mamba's global modeling capability.

To this end, we propose a locally sensitive SSM (LS-SSM). Unlike previous works applying Mamba directly as an encoder, our LS-SSM is designed with a local-to-global encoding strategy and fully exploits Mamba's selective feature extraction capacity. First, we incorporate the recently developed neighborhood attention (NATTEN) mechanism [24] to capture fine-grained local patterns and details. Then, an input discriminant enhancer (IDE) is designed to amplify feature distinctiveness across different spatial locations and channels and to generate more discriminative inputs for the following Mamba-based module [20] to selectively aggregate localized features. This composition results in a novel block capable of preserving detailed local information while modeling global dependencies. By stacking these blocks, we construct a hierarchical encoder that generates rich, multilevel feature representations. Note that the resulting features, which effectively capture both local and global information, are also well-suited for delineating building footprints.

To ensure effective decoupling of sharing encoder features for separate building height prediction and footprint segmentation within a unified MTL architecture, we extend VMamba's selective scanning mechanism [20] by reorienting it from the spatial dimension to the channel dimension, and design a channel-dimension selective SSM (CSSM) neck. This modification better reorganizes the shared encoder features and injects them into two task-specific decoding representations.

Cross-Task Feature Sharing Between Decoders: Task-specific information can be shared and exchanged again to enhance prediction performance across general vision-based tasks, as has been proven in [9], [27], and [28]. We also observed reports that cross-task footprint-height prediction interactions can yield mutual benefits [25], [26]. In these studies, features from task-specific encoders or decoders are fused by concatenation or addition [29], [30], and then recalibrated. However, this rough full-feature combination in previous studies does not seem reasonable enough. Sharing

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all features creates redundancy and adverse recoupling, and we believe that only sharing those mutually beneficial features can yield greater performance gains.

Motivated by this observation, we propose a K-top relevant feature mining (KTM) strategy to selectively leverage cross-task features and prune unnecessary parts. Specifically, learnable query embeddings are used to retrieve beneficial features for building height prediction and footprint segmentation from each other, via a cross-attention mechanism. The strategy encourages that only positively correlated features from one task are selected to assist the prediction of the other task through an auxiliary side prediction and supervision. This explicit mining mechanism not only enhances the performance of both tasks but also effectively prevents adverse intertask feature recoupling, providing greater benefits than the commonly used full feature sharing.

The main contributions of this study are summarized as follows.

1) We propose a LS-SSM that emphasizes the input-dependent nature of Mamba by prioritizing local feature representation prior to global selection. This design significantly enhances Mamba's suitability for RS data compared to its direct application as an encoder in current RS-related studies.
2) We introduce a KTM strategy to select beneficial cross-task features from task-specific decoders. This strategy facilitates more effective mutual reinforcement between building height prediction and footprint segmentation than previous full cross-task feature sharing.
3) We integrate LS-SSM, the KTM strategy, and the newly developed CSSM neck into a unified MTL network, termed the Mamba-based building height estimation network (MBHNet). MBHNet achieves state-of-the-art performance on two benchmark datasets, significantly outperforming recent MTL methods. This is also the first work to employ channel-dimension selective scanning for feature decoupling in a general MTL model.
4) There are a few large-scale building height maps, but with low geometric precision and even worse reliability. For example, the CMAB dataset [7] uses the number of floors to estimate height. High-precision building height maps are extremely lacking. We edit and release two high-resolution multitask datasets comprising optical imagery, DSMs, high-precision building footprints, and height annotations. These datasets facilitate diverse research in multimodal building height estimation, segmentation, and 3-D reconstruction.

II. Related Works

In this section, we review studies closely related to our work, including studies on building height estimation using RS data and MTL methods, as well as recent advancements in SSMs.

A. Building Height Estimation

1) Classic Building Height Estimation: To improve reliability, multiple types of RS data, such as optical imagery, SAR data, and DSM, have been utilized to predict heights and footprints of buildings. Wang et al. [10] proposed a hybrid multimodal fusion network that integrates Sentinel-1 SAR data with Sentinel-2 optical imagery for building height prediction; similar studies combining SAR and optical data can be found in [32]. Yu et al. [3] incorporated DSMs and orthophotos into a 3-D instance segmentation network to improve instance-level building height estimation accuracy.

There are also some monocular depth estimation studies originally raised from computer vision, featuring easy implementation but suffering from low reliability [33], [34], [35]. For example, [9], [14], [15] employed fully convolutional networks to learn both footprint segmentation and height estimation, and [33] and [36] estimate heights or the number of stories for each building from single RS images. Although suffering from significant prediction errors, this type of method is useful when only monocular images are available.

In this work, we adopt the DSM-optical image combined inputs to enhance reliability, investigate the potential of Mamba for building height mapping tasks, and adapt Mamba-based structures to RS data through a local-to-global encoding strategy.

2) Cross-Task Interaction in MTL: Previous research has demonstrated that cross-task interaction is critical for improving the performance of MTL [37]. This is typically implemented in decoder-focused methods, which share a general encoder and insert cross-task interactions among multiple decoders [38], [39]. Consequently, many studies have sought to enhance MTL models by reinforcing intertask feature correlation or sharing. For instance, [25] introduced a cross-task contrastive loss to maximize mutual information between features from different tasks, encouraging intertask consistency. Wang et al. [27] proposed an auxiliary boundary-attentive module to embed additional edge features and regularize the other two main tasks, i.e., semantic segmentation and height prediction. Chen et al. [35] developed an attention module to separate and fuse height and semantic features, improving multitask feature representation. Tang et al. [40] proposed a pixel-level interaction strategy and a logit-level interaction strategy to enhance the communication of information between change detection and semantic segmentation tasks. Shen et al. [9] and Lin et al. [39] introduced Mamba-based cross-task feature learning modules to model global intertask relationships.

The methods mentioned above are all effective; however, they all adopted the full-feature sharing strategy via convolution [27], [28], Transformer [25], or Mamba [9], [39]. In fact, not all features from one task are beneficial to the other task in an MTL framework; otherwise, why should we use more than one decoder?

To fix this issue, we propose a KTM strategy to produce cross-task prompts. This strategy dynamically filters out task-incompatible information and selects the K most positively relevant features. By selectively transferring cross-task features, our strategy brings significant mutual improvements to building height prediction and footprint segmentation tasks.

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B. State Space Models

SSMs describe the input–output relationship through a latent hidden state, and are originally derived from the classical Kalman filter and have been widely used in modern control systems [19]. In recent years, SSMs have been introduced into the field of natural language processing as an emerging DL architecture, demonstrating strong long-range sequence modeling capabilities with lower computational complexity compared to Transformer-based models [41], [42]. Mamba, a recent advanced variant of SSMs, integrates an input-dependent selection mechanism along with a hardware-efficient implementation, enabling impressive global modeling while maintaining linear computational complexity to sequence length [18]. To extend the use of Mamba to the 2-D vision domain, various adaptations have been proposed. For instance, Vim [43] introduces bidirectional scanning paths, while VMamba [20] further develops a four-way selective scanning strategy, allowing Mamba to generate effective visual representations for high-resolution images. Both Vim [43] and VMamba [20] implement spatial-dimension selective scanning.

Various Mamba-based variants have been explored in the RS field. To utilize Mamba's efficiency in global information modeling, He et al. [44] proposed Channel Swapping Mamba and Cross-Modal Mamba for pan-sharpening tasks. Chen et al. [45] introduced a multipath visual state space block within the RSMamba framework for RS image classification. Ma et al. [23] designed a dual-branch encoder that combines Mamba modules with convolutional operations to improve segmentation performance. Recent studies on employing Mamba to RS images can also be found in [22], [46], [47], [48], [49], [50], [51], and [52]. More closely related to our work, Mamba has also been applied to dense height prediction from single-view images [9], [39].

Although Mamba has been successfully applied in the RS field, its inherent adaptability for extracting building height information from aligned DSM and image data remains unexplored. Specifically, Mamba has thus far been used only in the feature decoding stage for height estimation [9], [39]. In addition, most RS studies have focused on Mamba's global modeling capabilities, with limited attention given to its input-dependent characteristics. This gap highlights both the necessity and the motivation for our study.

III. Methods

A. Preliminaries

State Space Model: SSM originates from the classic Kalman filter and linear time-invariant system theory; it maps an input sequence $x(t) \in R$ to an output sequence $y(t) \in R$ via a hidden state $h \in R^N$ [19], [20]. Mathematically, the continuous-time SSM can be formulated using linear ordinary differential equations

$$h'(t) = Ah(t) + Bx(t) \tag{1}$$

$$y(t) = C^{\top} h(t) \tag{2}$$

where $A \in R^{N\times N}$ denotes the state matrix, $B \in R^N$ refers to the input matrix, $C \in R^N$ represents the output matrix, the symbol "T" denotes the matrix transpose operation, and $h(t)$ is the hidden state at time t. Equation (1) describes the evolution of the hidden state $h(t)$, while (2) produces the output as a linear transformation of the hidden state $h(t)$.

Discretization of SSM: To integrate SSM into DL models, a discretization procedure is required. A widely adopted way is the zero-order hold approach, which converts the continuous matrices A and B to their discretized counterparts with a time step parameter, denoted as

$$\bar{A} = \exp(\Delta A) \tag{3}$$

$$\bar{B} = (\Delta A)^{-1}\left(\exp(\Delta A) - I\right) \cdot (\Delta B) \tag{4}$$

where I is the identity matrix. Based on this transformation, the discretized forms of (1) and (2) can be expressed as

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t \tag{5}$$

$$y(t) = \bar{C}^{\top} h(t). \tag{6}$$

Mamba: Structured state space sequence model (S4) [19] treats A, B, C, and in (3)~(6) as trainable parameters, and Mamba [18] further introduces the selective scanning mechanism built on S4 (abbreviated as S6), where B, C, and are modeled as functions of the input x_t. S6 allows Mamba to selectively retain relevant information and suppress irrelevant content based on both current and historical input. The output of Mamba can be computed via a global convolution, as follows:

$$\bar{K} = \left(\bar{C}\bar{B}, \bar{C}\bar{A}\bar{B}, \ldots, \bar{C}\bar{A}^{L-1}\bar{B}\right) \tag{7}$$

$$y = x * \bar{K} \tag{8}$$

where $\bar{K} \in R^L$ is the structured convolutional kernel, and L is the length of the input sequence. Therefore, Mamba represents a selective SSM that integrates dynamic selection mechanisms into the state update process.

B. Overview of MBHNet

The overall network architecture of the proposed MBHNet is illustrated in Fig. 2. MBHNet adopts an MTL framework, taking aligned orthophotos and DSMs as its input, and producing pixel-wise building footprint and height maps simultaneously. MBHNet consists of four key components: 1) a LS-SSM encoder for data-customized feature extraction; 2) a semantic decoder for building footprint segmentation; 3) a height decoder for building height estimation, both including a CSSM-based bottleneck and a series of fusion and upsampling blocks; and 4) a KTM module to enable effective cross-task interactions. To guide KTM in mining beneficial features for different tasks, auxiliary output supervision is applied to both decoders during training. During inference, height predictions outside the predicted building footprints are masked out to align with semantic boundaries and height. In Sections III-C–III-G, we detail each component and the associated loss functions.

C. LS-SSM Encoder

We design the local-then-global LS-SSM encoder, as illustrated in Fig. 3, for extracting adequate features from complex

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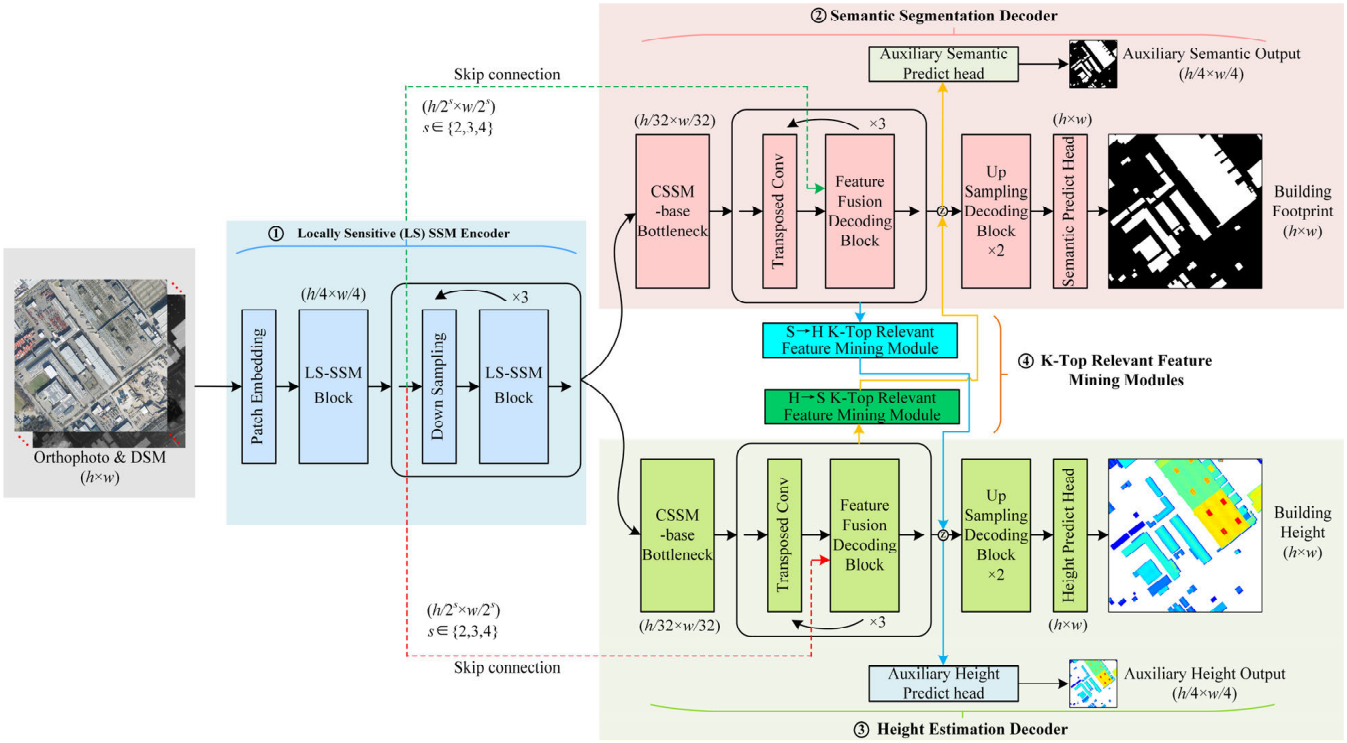


Fig. 2. Architecture of the proposed Mamba-based fine-grained building height estimation network, referred to as MBHNet. Circled z means feature fusion operation via feature concatenation and a depth-wise separable convolution.

RS data. This encoder is composed of the following three core components.

1) Patch embedding transforms the input into initial feature representations. Patch embedding first divides the input into nonoverlapping patches with a window size of 4×4. Each patch is processed by a linear projection layer to generate fixed-dimensional feature vectors (96 dimensions in our case).
2) LS-SSM block comprises three parts: a local focusing subblock, a global modeling subblock, and a linear transform layer, as detailed in Figs. 3 and 4. To construct the local focusing subblock, we first introduce the recently developed NATTEN mechanism, NATTEN [24], which performs self-attention within a fixed-size local window for each pixel using a sliding window approach, effectively capturing fine-grained local patterns and details from RS images.

Subsequently, the Mamba-based global modeling subblock is applied to selectively aggregate local information. Unlike methods that directly adopt standard Mamba or VMamba blocks [23], [50], we incorporate an IDE to improve the original Mamba's input-dependent selection capability. As illustrated in Fig. 3(b), IDE is a new module we designed to enhance input feature discrimination across both channel and spatial dimensions. Given the input feature map $F \in R^{c\times h\times w}$, IDE first computes the variance for each feature channel to obtain $V \in R^{c\times 1\times 1}$, and then computes the mean of each pixel across channels to obtain $M \in R^{1\times h\times w}$. Higher variance indicates greater spatial variation and richer details, and thus stronger channel importance [53], while pixels with high responses across multiple channels yield higher mean values, highlighting spatial importance. To recalibrate these importance values, a fully connected (FC) layer is applied to the channel-wise variance matrix, and a 7×7 convolution is applied to the pixel-wise mean matrix. The FC layer preserves the channel dimensionality ($c \times 1 \times 1 \rightarrow c \times 1 \times 1$), and the single-channel convolution layer preserves spatial resolution ($1 \times h \times w$). Both outputs are activated by sigmoid functions. Their element-wise product is then further normalized via another sigmoid to produce a unified importance map $S \in R^{c\times h\times w}$. Finally, a gating switch converts this importance into gating weights as follows:

$$w_{i,j,k} = \begin{cases} S_{i,j,k}, & if\ S_{i,j,k} < T \\ 1, & \text{otherwise} \end{cases} \tag{9}$$

where $S_{i,j,k}$ is the importance score at a specific 3-D feature map position of (i, j, k), $w_{i,j,k}$ is the corresponding gating weight, and $T = 0.5$ is the fixed threshold to distinguish informative and less informative responses, following [53]. By applying these gating weights to the input features, IDE explicitly amplifies features' discrimination for specific tasks.

As illustrated in Fig. 3(a), the global modeling subblock comprises two branches. The main (bottom) branch includes a layer normalization, a linear layer followed by depth-wise convolution and a SiLU activation for feature dimension expansion, an IDE module for enhancing feature discrimination, a spatial-dimension selective SSM module (SS-SSM) to capture global dependencies, and a final layer normalization. Specifically, we adopt the spatial-dimension selective SSM (SS2D) module in VMamba [20] as the SS-SSM, as illustrated

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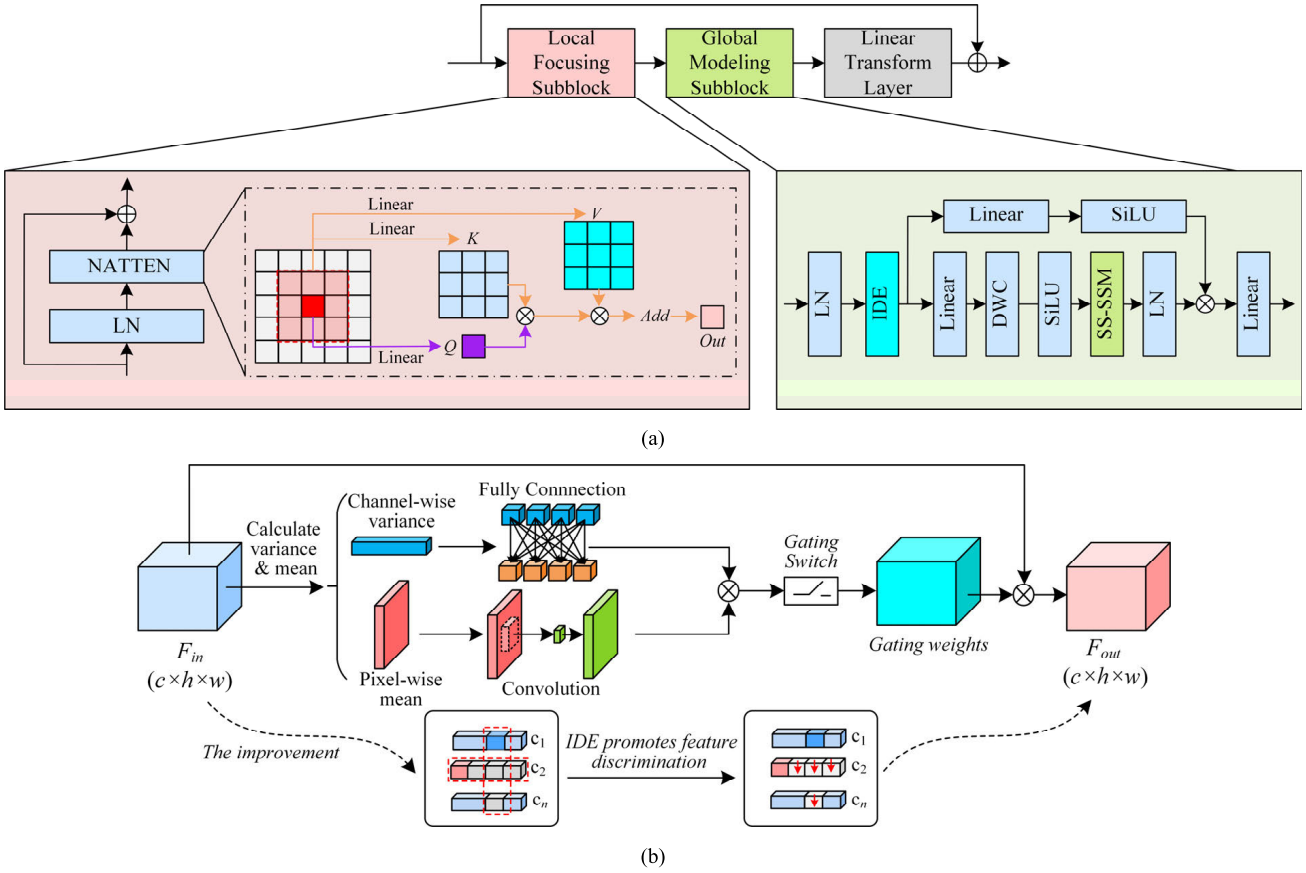


Fig. 3. (a) LS-SSM block, consisting of a local focusing subblock, a global modeling subblock, and a linear transformation layer. (b) Detailed structure of the proposed IDE.

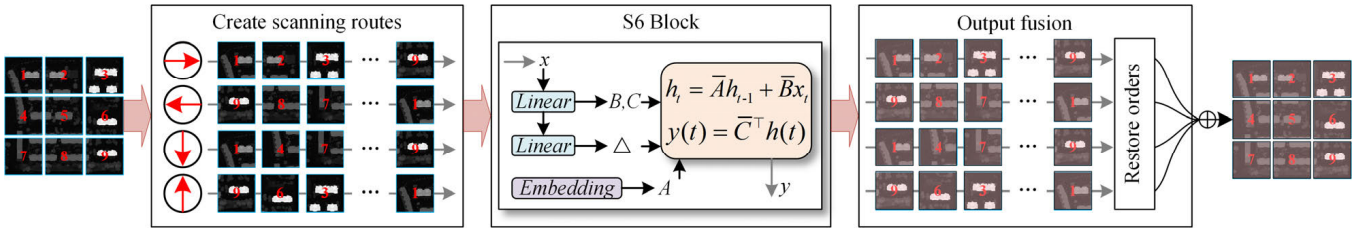


Fig. 4. Illustration of the SS2D, which adapts the S6 architecture for image processing tasks by performing 2-D selective scanning.

in Fig. 4, which adapts the S6 architecture for 2-D visual tasks. The top branch provides an input-dependent gating mechanism consisting of a linear layer and a SiLU activation.

The linear transform at the end of the LS-SSM block is straightforward, consisting of a linear layer followed by a SiLU activation. The top residual connection in the LS-SSM block is introduced to ease the learning process.

1) Downsampling in the LS-SSM encoder is performed via patch merging operations [54], which halve the feature map size and double the channel count.

By stacking three groups of downsampling layers and LS-SSM blocks sequentially, LS-SSM creates hierarchical feature representations with strides of 1/4, 1/8, 1/16, and 1/32 relative to the original input. It should be emphasized that both the introduced NATTEN and the newly designed IDE are crucial to adapting the Mamba-based encoder to RS image feature extraction, and their integration significantly outperforms directly using the very well-known VMamba [20], as demonstrated in the subsequent ablation study.

D. Semantic Decoder and Height Decoder

The semantic decoder and height decoder share the same architecture except for their final output layers. As shown in Fig. 2, a CSSM-based bottleneck first transforms the encoding features into decoding features, acting as a bridge between the encoder and decoders. Subsequently, a transposed convolution and a feature fusion decoding block are applied three times to restore the features to 1/4 of the original input resolution. Each feature fusion decoding block receives both the upsampled features from the transposed convolution and the corresponding skip connection features from the encoding stage. After

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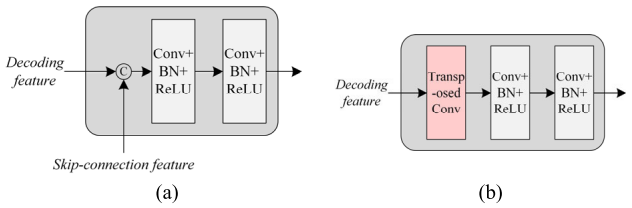


Fig. 5. (a) Feature fusion decoding block and (b) upsampling decoding block.

that, the output is passed to the KTM module for positively relevant cross-task feature mining, then fused with mined features from the other task to serve as updated features for the next decoding stages. Finally, two groups of upsampling decoding blocks further restore the features to the original input resolution, and the task-specific prediction head maps the final feature to the desired output. Specifically, the height prediction head predicts absolute building height values from the concatenation of the DSM and high-level decoding features (the concatenation is omitted in Fig. 2 for clarity), whereas the semantic prediction head predicts per-pixel scores from the decoding features, followed by a sigmoid activation and a threshold of 0.5 to generate binary segmentation. Both the feature fusion decoding block and the upsampling decoding block are detailed in Fig. 5.

E. CSSM-Based Bottleneck

The SS2D [20] employed in the encoding stage captures global dependence along the spatial dimension. Four scanning routes are constructed, and the outputs from all routes are fused to share information from different directions. As illustrated in Fig. 6(a), each dashed box represents an element on the scanning paths. Selective scanning along the spatial dimensions effectively aggregates spatial contextual information. However, to decouple task-aware features from shared encoder features, the relationships among different channels are more critical than those along the spatial dimension. Previous channel attention mechanisms [55] excessively compress spatial dimensions through global pooling to produce channel descriptors. The compressed descriptors may harm the following interchannel relationship modeling and feature recalibration.

A natural extension is to model interchannel relationships using another selective SSM. This approach is conceptually analogous to spectral scanning strategies employed to capture category cues from numerous spectral bands [47], [52]. To this end, we design the CSSM-based bottleneck by modifying the spatial-dimension selective SSM into a CSSM, differing primarily in their scanning routes. To construct channel-dimension scanning paths, all spatial pixels within each channel are flattened into one dimension. Two scanning directions are then established along the channel dimension: one from front to back and the other from back to front, as illustrated in Fig. 6(b). This design preserves the full spatial information without any compression, and the resulting CSSM bottleneck is depicted in Fig. 6(c).

F. KTM Module

Task-specific decoders decouple the shared encoder features into task-aware representations under independent supervision signals, generating customized model parameters for each task. Recently, intertask interactions have proved effective [9], [27], [28], [37], [39]. However, these studies have commonly adopted full feature sharing between tasks. Full feature sharing introduces redundancy, and the resulting recoupling may adversely affect prediction performance. We propose the KTM module, which enables the explicit mining of mutually beneficial features between tasks. As illustrated in Fig. 7(a), the KTM module determines feature correlations through a self-attention mechanism and a cross-attention mechanism, and selects a specific proportion of features from one task to support the prediction of the other task.

Taking the transfer of features from the height prediction task to the semantic segmentation task ($h \rightarrow s$) as an example, k query embeddings (Q) are first initialized using a truncated normal distribution and then refined through a self-attention operation. Inspired by DETR [31], the self-attention mechanism enables each query embedding to access information from all other queries, thereby actively avoiding duplicate queries. Features from the height estimation task F_{task-h} are projected into a key vector (K) and a value vector (V) through two parallel linear layers. Unlike the self-attention operation, where queries, keys, and values are derived from the same input, the cross-attention operation computes attention weights between inputs from different sources, enabling specific information integration based on the context provided by the queries.

To align the sequence length between the key feature $K \in R^{c\times h\times w}$ and the query $Q \in R^{k\times r}$, we first flatten the feature K along the spatial dimension, resulting in $K' \in R^{c\times hw}$. A FC layer is then applied to squeeze the spatial dimension from $h \times w$ to r ($r = 512$ in this case), yielding $K'' \in R^{c\times r}$. A compatibility function [56] followed by a softmax operation is used to compute the correlations between K'' and Q, producing task-relevant scores $S \in R^{k\times c}$. Finally, the channels from V corresponding to the k highest scores in S are selected to support the footprint segmentation. The KTM process is formalized as:

$$K,\ V = \text{Linear}\left(F_{task-h}\right) \tag{10}$$

$$K'' = \text{FC}\left(\text{Flatten}\left(K\right)\right) \tag{11}$$

$$S = \text{Soft max}\left(K''Q\right) \tag{12}$$

$$F_{h\rightarrow s} = \Omega_{k-top}\left(V, S\right) \tag{13}$$

where Ω_{k-top} denotes the selection operator that retains the channels in V corresponding to the k top scores in S.

As illustrated in Fig. 7, the relevant feature $F_{h\rightarrow s}$ is integrated with the original features in the semantic decoder to generate the auxiliary side semantic prediction and the final prediction. This design encourages that positively contributing features from the height prediction task are transferred to the semantic segmentation task. Just a subset of task-specific features is shared, thereby suppressing feature redundancy and preventing adverse recoupling between tasks. The same principle applies to the feature mining process from the

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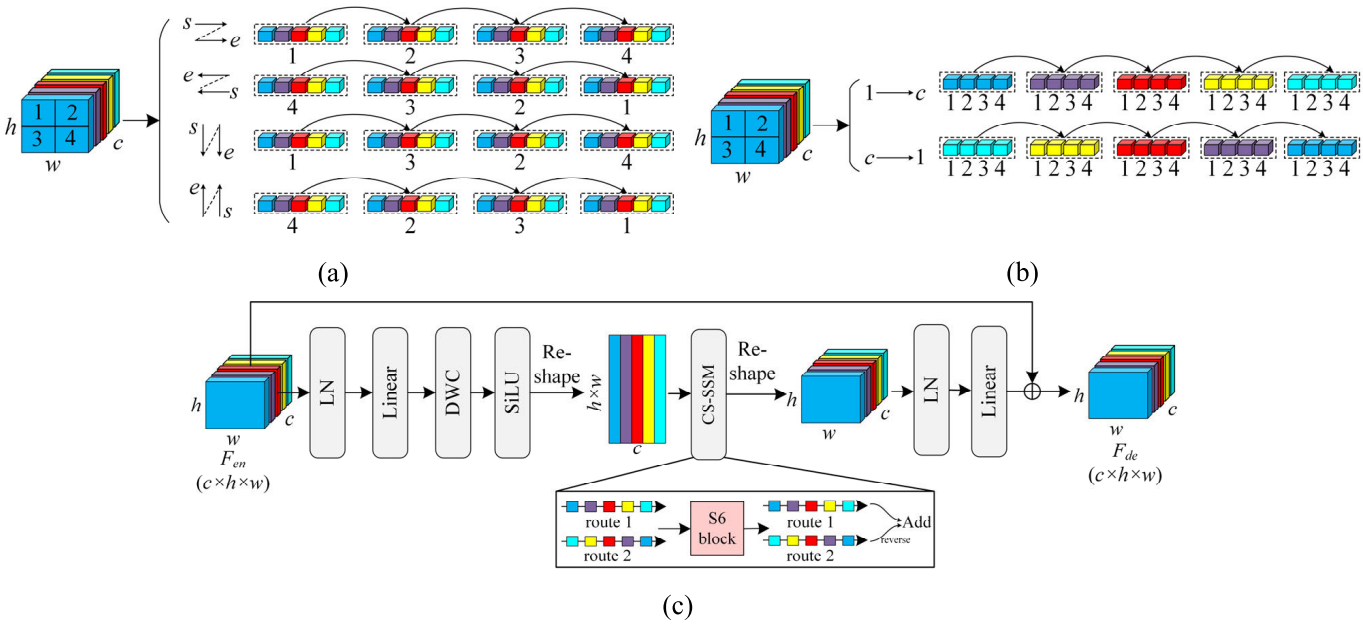


Fig. 6. (a) Spatial-dimension four-way scanning routes. (b) Channel-dimension bi-directional scanning routes. (c) CSSM-based bottleneck. "s" and "e" denote the start and end positions, while "c," "h," and "w" represent the number of channels, height, and width of the feature map, respectively.

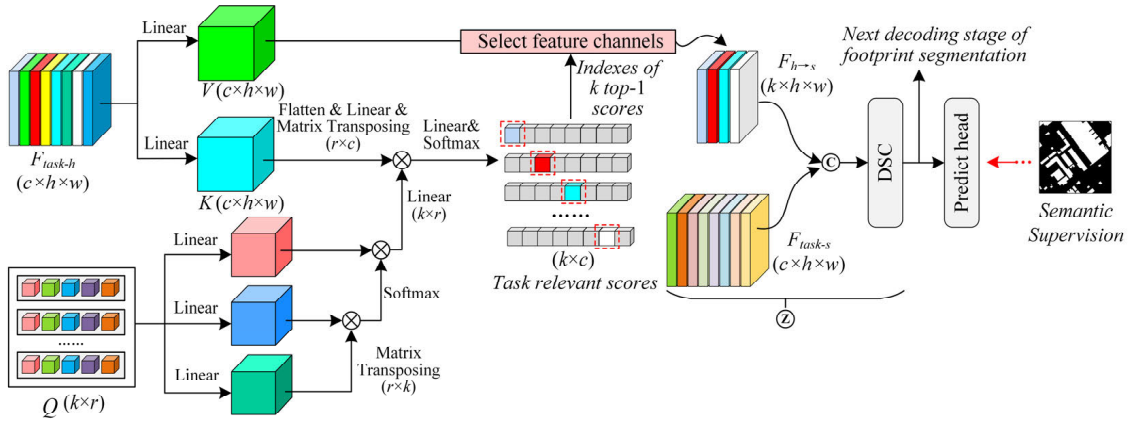


Fig. 7. KTM module, using the $h \rightarrow$ s feature transfer as an example. The mined features are fused into the decoding features of the segmentation task via feature concatenation and a depth-wise separable convolution (denoted by the symbol ⓩ in Fig. 2). The fused features serve as updated decoding features for producing the auxiliary prediction through a side prediction head.

semantic segmentation branch to the height prediction branch, which is not elaborated here for brevity.

G. Loss Functions

To optimize the learning of the two tasks embedded in the proposed MBHNet, a multitask joint learning loss is applied. Following prior studies [15], [25], [28], we adopt the widely used mean absolute error (MAE) loss for height estimation. The MAE loss is defined as

$$\mathcal{L}_{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|h_i - \hat{h}_i\right| \tag{14}$$

where $\hat{h}_i$ enotes the predicted building height value for pixel i, h_i is the corresponding ground truth (GT), and n is the total number of building pixels.

Meanwhile, binary cross-entropy (BCE) loss and Dice loss [57] are applied to the building footprint segmentation task, offering complementary objectives. BCE provides pixel-wise supervision, while Dice loss captures global segmentation performance by evaluating the overlap between prediction and GT. The BCE loss is defined as

$$\mathcal{L}_{BCE} = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log\left(\hat{y}_i\right) + \left(1 - y_i\right)\log\left(1 - \hat{y}_i\right)\right] \tag{15}$$

where $\hat{y}_i$ denotes the predicted probability value of pixel i belonging to buildings, which has been normalized by a sigmoid function. y_i is the corresponding GT category, 1 is for buildings, and 0 is for others. The Dice loss is defined as

$$\mathcal{L}_{DICE} = 1 - \frac{2\sum_{i}^{N} y_i \hat{y}_i}{\sum_{i=1}^{N} y_i^2 + \sum_{i=1}^{N} \hat{y}_i^2} \tag{16}$$

where N is the total number of the whole prediction map, and $N > n$.

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To jointly supervise the low-resolution side predictions and the final fine-grained predictions, the overall multitask loss function is defined as

$$\mathcal{L}_{total} = \alpha \mathcal{L}_{final} + \beta \mathcal{L}_{side} \tag{17}$$

$$\mathcal{L}_{final}\,(\text{or}\,\mathcal{L}_{side}) = \mathcal{L}_{MAE} + \frac{1}{2}\left(\mathcal{L}_{BCE} + \mathcal{L}_{DICE}\right) \tag{18}$$

where $\alpha = 1$ and $\beta = 0.05$, experimentally determined through an ablation study in Section IV-D. $\mathcal{L}_{final}$ and $\mathcal{L}_{side}$ share the same combination of loss function components, but are applied at different outputs.

IV. Experiments and Results

A. Datasets

1) GeoNRW Dataset: The original GeoNRW dataset [58] includes orthorectified aerial photographs [each with a resolution of 1000 × 1000 pixels at 1-m ground sampling distance (GSD)], co-registered DSMs derived from LiDAR, and GT land cover maps containing ten classes. To support our studies on building semantic segmentation and height prediction, we reprocessed the original dataset as follows. First, we cropped the large orthophotos, DSMs, and GT maps into 512 × 512-pixel tiles. To alleviate class imbalance, we excluded tiles in which building pixels accounted for less than 10%. Next, the GT land cover maps were reclassified into two categories, i.e., building and background, yielding binary semantic labels. GT building height maps were generated using a semi-automatic procedure. Specifically, DTMs were generated using an automatic filtering algorithm that combines adaptive local terrain filtering with neighborhood buffer analysis [12], where the filtering window size is determined according to the real building sizes. For detailed implementation procedures, we refer readers to [12]. This automated process was subsequently refined through manual inspection and correction to ensure accuracy. Each individual building instance in the DTM was assigned a uniform base elevation. Finally, the pixel-wise GT building heights were computed by subtracting the DTM from the DSM within the building footprints. The reprocessed GeoNRW dataset was randomly split into training, validation, and test sets with a 50%:25%:25% ratio, resulting in 2896 training tiles, 1448 validation tiles, and 1448 test tiles, a total coverage area of approximately 1500 km^2.

2) Tianjin Dataset: Tianjin dataset consists of high-resolution (0.2 m) aerial orthophotos and DSMs generated from aerial stereo photogrammetry, covering Xiqing, Hebei, and Heping Districts of Tianjin, China. The GT building height maps were produced following the same procedure as for the GeoNRW dataset. DTMs were generated via terrain-adaptive filtering, manually refined, and height values were computed as the difference between the DSM and DTM within building footprints. Data from Xiqing District was divided into three subsets: a training set, a validation set, and Test Set-I. Data from the Hebei and Heping Districts was used as Test Set-II to evaluate the model's generalization ability. The training set covers approximately 1.30 km^2, and the validation set spans 0.49 km^2. Test Set-I and II cover 0.88 and 0.56 km^2, respectively. For efficient model training, the orthophotos and DSMs in the training and validation sets were also cropped into 512 × 512-pixel tiles. The test images remain uncropped; instead, the DL models dynamically load and process regions from the large test images online, generate predictions, and merge the outputs.

Fig. 8 presents examples of orthophotos and corresponding GT building height maps from the GeoNRW dataset, while Fig. 9 provides an overview of the dataset splits for Tianjin dataset. In both datasets, structural differences among building roofs are clearly captured through pixel-level height variations. Note that for visualization, building heights in each image were individually normalized before color mapping. Normalization across all images was avoided to preserve the visualization of fine-scale height variations within each individual scene.

3) Potsdam Dataset: Potsdam dataset[1] is a widely used benchmark for dense height estimation. It consists of 38 very high-resolution (0.05 m) aerial orthophoto images, semantic labels, DSMs, and nDSMs, covering a historic city in Germany. Each image/DSM/nDSM/label map has a spatial size of 6000 × 6000 pixels. To adapt this dataset to our building height mapping task, we convert the six-class semantic labels into a binary mask containing only buildings and background. All nonbuilding regions in the nDSM are then set to zero. Following [9], the red, green, and blue bands of the imagery are used in our experiments, and the images with IDs 2_10, 2_11, 2_12, 3_10, 3_11, 3_12, 4_10, 4_11, 4_12, 5_10, 5_11, 5_12, 6_7, 6_8, 6_9, 6_10, 6_11, 6_12, 7_7, 7_8, 7_9, 7_10, 7_11, and 7_12 for training and the remaining 14 images for testing.

Both the training and test data were downsampled to 0.2 m to avoid cases where building roofs occupy entire tiles and to retain ground reference. Each training image, DSM, nDSM, and binary mask are then cropped into 512 × 512-pixel tiles for model training. The test data is not cropped offline, and the inference procedure remains consistent with that used for Tianjin dataset.

B. Evaluation Metrics and Implementation Details

1) Evaluation Metrics: To comprehensively evaluate the performance of the proposed method, we report metrics for both building height prediction and footprint segmentation. For height prediction, we adopt three commonly used evaluation indicators, including MAE, root-mean-square error (RMSE), and absolute relative error (absRel), following previous works [9], [10], [25], [26]. The formulations of these metrics are as follows:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|h_i - \hat{h}_i\right| \tag{19}$$

$$\text{RMAE} = \sqrt{\frac{1}{n}\sum\nolimits_{i=1}^{n}\left(h_i - \hat{h}_i\right)^2} \tag{20}$$

$$\text{absRel} = \frac{1}{n}\sum_{i=1}^{n}\left|h_i - \hat{h}_i\right|/h_i \tag{21}$$

[1] https://www.isprs.org/education/benchmarks/UrbanSemLab/

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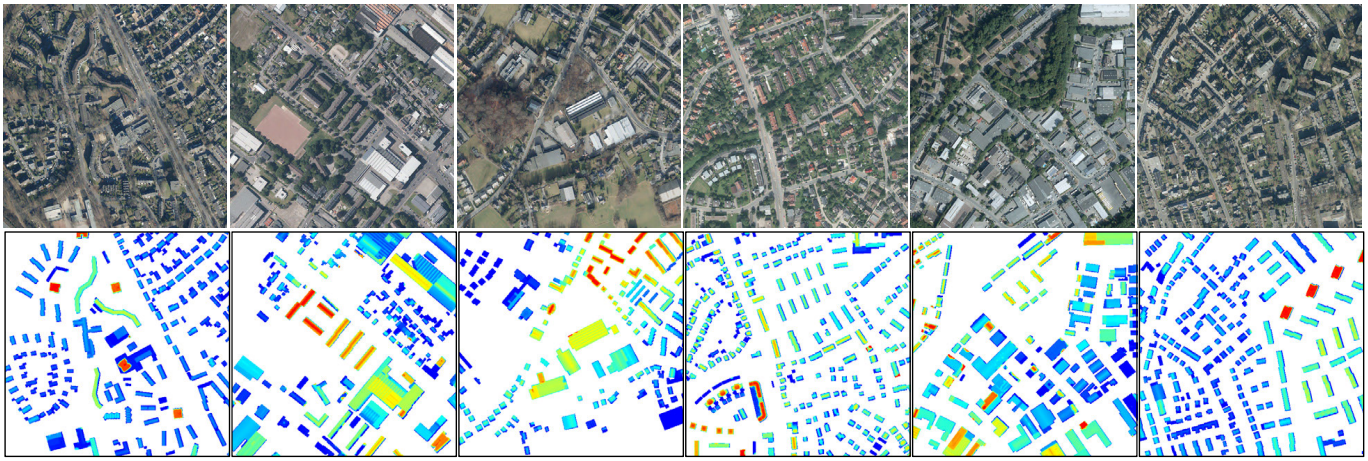
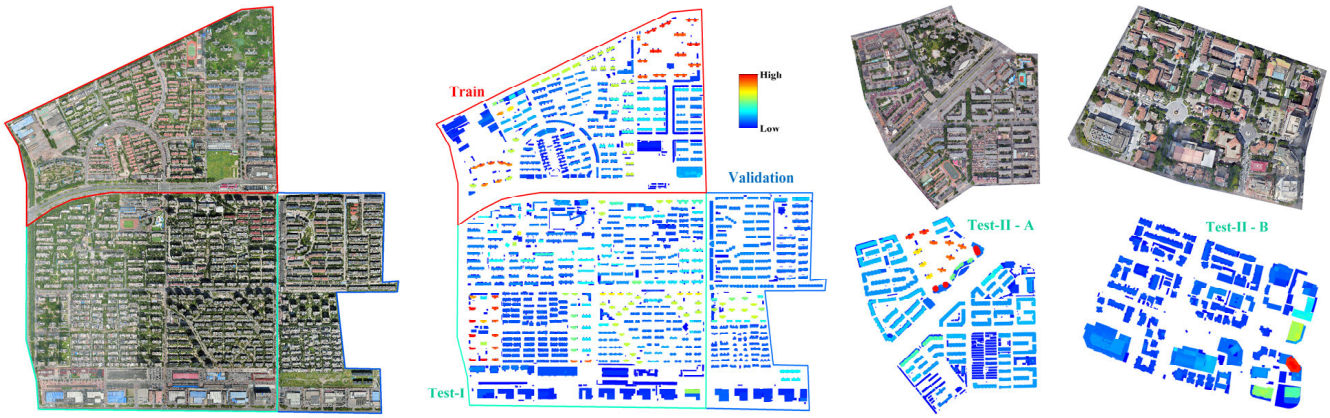


Fig. 8. Examples of orthophotos (the top row) and their corresponding pixel-wise building height maps (the bottom row) in the reprocessed GeoNRW dataset.



(a) Tianjin Xiqing District (b) Tianjin Hebei District (c) Tianjin Heping District

Fig. 9. Overview of orthophotos and their corresponding pixel-wise building height maps of Tianjin dataset. (a) Tianjin Xiqing District. (b) Tianjin Hebei District. (c) Tianjin Heping District.

where n denotes the number of building pixels, $\hat{h}_i$ is the predicted building height for pixel i, and h_i is the corresponding GT.

For footprint segmentation, three widely used evaluation metrics are adopted, including intersection over union (IoU), $F1$-score, and overall accuracy (OA). These metrics are defined as follows:

$$\text{IoU} = \frac{\|\text{TP}\|}{\|\text{TP}\| + \|\text{FP}\| + \|\text{FN}\|} \tag{22}$$

$$\text{F1} = \frac{2 \cdot \|TP\|}{2 \cdot \|TP\| + \|FP\| + \|FN\|} \tag{23}$$

$$\text{OA} = \frac{\|TP\| + \|TN\|}{\|TP\| + \|TN\| + \|FP\| + \|FN\|} \tag{24}$$

where TP, FP, FN, and TN represent the predicted pixels classified as true positive, false positive, false negative, and true negative. The $\|*\|$ symbol denotes the count of the corresponding pixel category. Note that IoU and $F1$-score are only calculated for the building category, while OA is computed over all pixels.

2) Implementation Details: To ensure fair comparisons, we reimplemented all methods using the PyTorch DL framework on the same Windows workstation, which was equipped with an NVIDIA GeForce RTX 4090D GPU, a 13th-generation Intel[2] Core[3] i7-13700 CPU, and 64 GB of RAM. The maximum number of training epochs was set to 60 for the GeoNRW dataset, 200 for Potsdam dataset, and 400 for Tianjin dataset. All backbone networks were initialized with ImageNet-pretrained weights [59]. The initial learning rate was set to $1e^{-4}$, and model training was conducted using the AdamW optimizer with a cosine learning rate annealing schedule. Data augmentation strategies, including random color jittering, horizontal and vertical flipping, were uniformly applied during training for all methods but were disabled during testing.

The inputs, consisting of orthophotos and DSMs, were normalized following the preprocessing procedure described in [3]. Since DSM encodes relative height variations across building roofs, it was reintegrated into all height prediction heads in both our method and the comparison methods throughout training and inference by concatenating it with the decoder features. This straightforward reuse reduces the complexity of height learning and improves training stability.

[2]Registered trademark.
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TABLE I
QUANTITATIVE RESULTS OF DIFFERENT MTL METHODS ON THE GEONRW DATASET

Method	Height Estimation			Footprint Segmentation			Inference Time (seconds)
	MAE (m) ↓	RMSE (m) ↓	absRel ↓	IoU (%) ↑	F1 (%) ↑	OA (%) ↑	
Method by (*Srivastava et al.*, 2017)	1.450	3.298	0.140	72.3	83.9	94.9	45.01
Method by (*Carvalho et al.*, 2020)	1.268	2.353	0.127	77.2	87.2	95.9	**39.90**
BAMTL (Wang *et al.*, 2021)	0.963	1.997	0.099	73.2	84.5	95.1	51.87
SCE-Net (*Xing et al.*, 2022)	0.966	2.015	0.095	76.6	86.7	95.8	55.43
MFTSC (*Chen et al.*, 2023)	1.103	2.300	0.110	76.4	86.6	95.7	42.17
MTMamba (*Lin et al.*, 2024)	0.733	2.201	0.068	74.2	85.2	95.2	67.01
RSMTMamba (*Shen et al.*, 2025)	0.822	2.276	0.078	77.7	87.4	95.9	47.70
MBHNet (Ours)	**0.640**	**1.881**	**0.064**	**79.5**	**88.6**	**96.3**	45.17

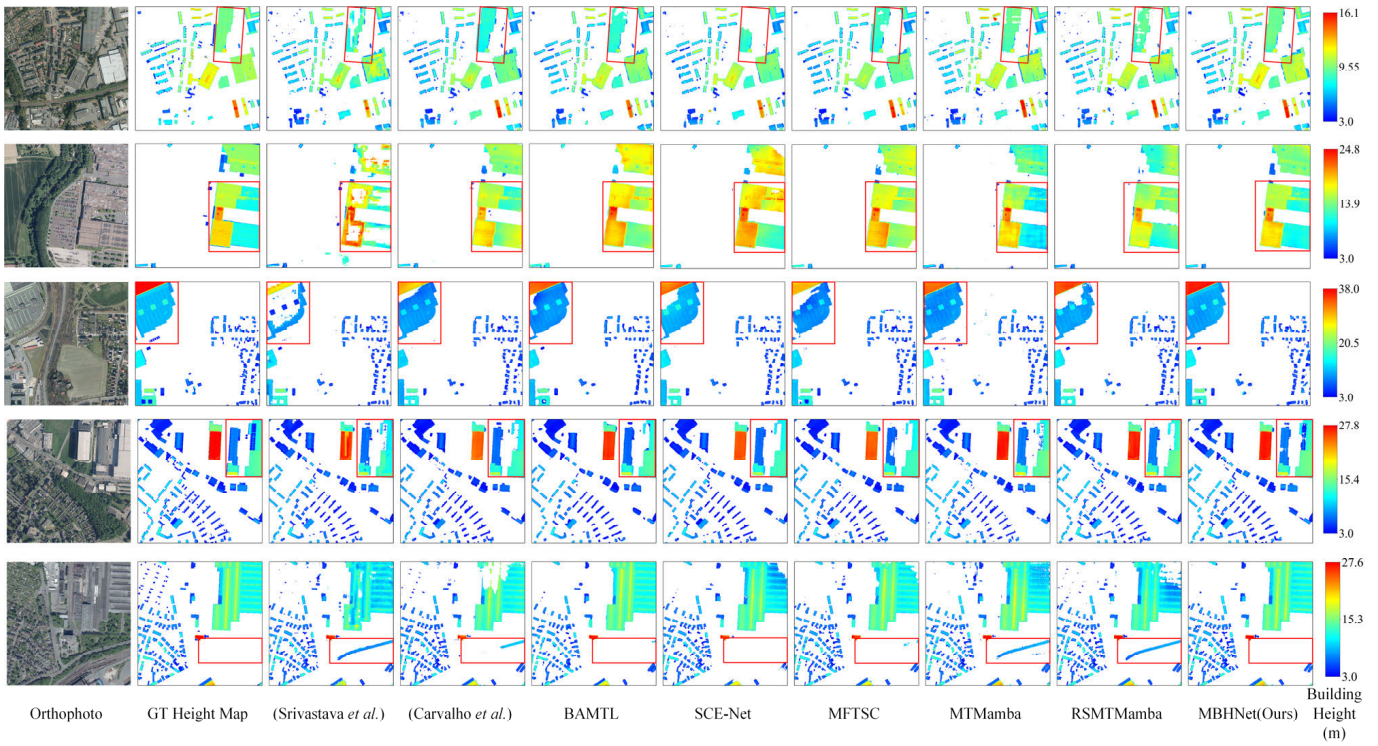


Fig. 10. Visualization of predicted building height maps generated by different comparison methods on the GeoNRW dataset.

C. Comparison With State-of-the-Art Methods

We compared the proposed MBHNet with other recent MTL-based height estimation methods, including the methods developed by Srivastava et al. [14] and Carvalho et al. [15], BAMTL [27], SCE-Net [28], MFTSC [26], MTMamba [39], and RSMTMamba [9]. These comparison methods, except for MTMamba, are all from the RS community. The method proposed by Srivastava et al. [14] adopted a simple architecture by stacking multiple convolution and pooling layers for feature extraction. Carvalho et al. [15] employed DenseNet [60] as the backbone. In contrast, SCE-Net, BAMTL, and RSMTMamba utilized ResNet [61] as their encoder backbone, while MFTSC and MTMamba adopted the Swin Transformer [54] for feature encoding. As mentioned above, Mamba-based encoders have not been employed in the height estimation from RS data.

1) GeoNRW Dataset: Table I presents the quantitative results of all methods evaluated on the GeoNRW dataset. The proposed MBHNet achieves the best performance on both building height prediction and footprint segmentation tasks. For height prediction, MBHNet surpasses the second-best method, MTMamba [39], by an absolute margin of 0.093 m (a relative improvement of 12.7%) in terms of MAE. Regarding footprint segmentation, MBHNet outperforms the second-best method, RSMTMamba [9], by 1.8% in IoU, 1.2% in $F1$-score, and 0.4% in OA, respectively. The consistent improvements in both geometric and semantic tasks demonstrate the effectiveness of our proposed method, in which the LS-SSM encoder (for data-customized feature encoding), the channel-dimension selective SSM bottleneck (for task-adaptive feature decoupling), and the KTM module (for selective cross-task feature prompting rather than full feature sharing) collaborate synergistically. These design concepts remain underexplored in the existing comparison methods. Another observation is that methods incorporating Mamba-based structures, namely, MBHNet, MTMamba [39], and RSMTMamba [9], generally outperform other methods overall, highlighting Mamba's superiority in capturing long-range dependencies for height

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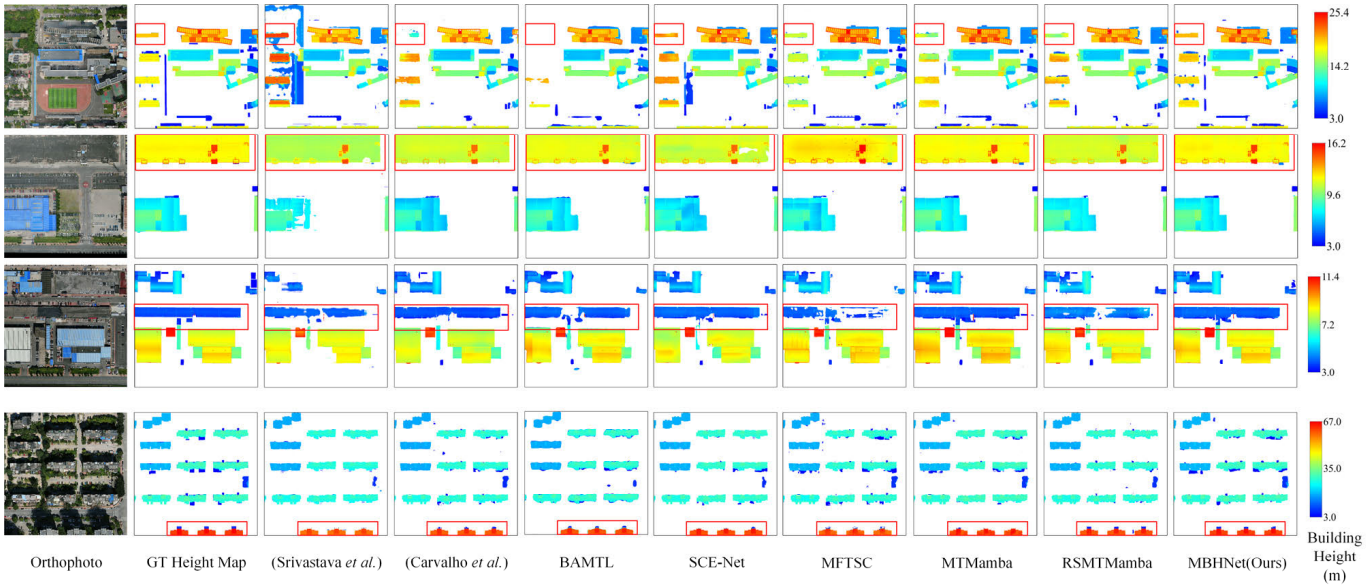


Fig. 11. Visualization of predicted building height maps generated by different comparison methods on the Test Set-I of Tianjin dataset.

TABLE II
QUANTITATIVE RESULTS OF DIFFERENT MTL METHODS ON THE TEST SET-I OF TIANJIN DATASET

Method	Height Estimation			Footprint Segmentation			Inference Time (seconds)
	MAE (m) ↓	RMSE (m) ↓	absRel ↓	IoU (%) ↑	F1 (%) ↑	OA (%) ↑	
Method by (*Srivastava et al.*, 2017)	1.485	2.617	0.095	76.2	86.5	93.5	7.49
Method by (*Carvalho et al.*, 2020)	0.989	1.356	0.057	86.8	92.9	96.7	**6.64**
BAMTL (Wang *et al.*, 2021)	0.628	1.279	0.044	84.0	91.3	96.0	8.60
SCE-Net (*Xing et al.*, 2022)	0.716	1.264	0.044	86.4	92.7	96.6	9.85
MFTSC (*Chen et al.*, 2023)	0.812	1.527	0.050	84.5	91.6	96.0	6.93
MTMamba (*Lin et al.*, 2024)	0.437	0.956	0.026	87.1	93.1	96.7	11.10
RSMTMamba (*Shen et al.*, 2025)	0.689	1.067	0.042	87.1	93.1	96.8	7.88
MBHNet (Ours)	**0.294**	**0.565**	**0.022**	**88.2**	**93.7**	**97.0**	7.65

estimation from RS data. Apart from the two earlier methods [14], [15], MFTSC [26] performs worse than the others since it is the only method without explicit cross-task feature interaction.

Several qualitative comparison examples of predicted building height maps from the GeoNRW dataset are presented in Fig. 10. Distinguishable regions are highlighted with red rectangular boxes to facilitate a clearer comparison among different methods. As shown in the first two rows, our method consistently produces color patterns more closely aligned with the GT height maps at the same locations than others, indicating higher height estimation accuracy. Meanwhile, our method also demonstrates superior building segmentation quality in terms of both completeness and correctness. For instance, the large building in the third row and the inner regions of buildings in the fourth row predicted by our method exhibit fewer missing segments than those predicted by other methods. Furthermore, as illustrated in the fifth row, some methods, such as MTMamba [39] and RSMTMamba [9], incorrectly classify trains on the railway as buildings. In contrast, our method yields fewer misclassification errors in such scenarios.

2) Tianjin Dataset (Test-I): Table II presents the quantitative results on Test Set-I of Tianjin dataset. The first observation is that the higher spatial resolution (0.2 m) of Tianjin dataset leads to better overall performance in both building height prediction and footprint segmentation across all methods compared to the GeoNRW dataset (1 m), indicating the value of our proposed Tianjin dataset to building height mapping research. The other conclusions are consistent with those drawn from the GeoNRW dataset. Specifically, our proposed method achieves the best performance on both tasks. MTMamba [39] ranks second in building height prediction, while RSMTMamba [9] achieves the second-best performance in footprint segmentation. MFTSC [26] underperforms compared to the methods that explicitly incorporate cross-task feature interaction. The significant reduction in MAE and RMSE from 0.437 to 0.956 m achieved by the second-best method to 0.294 and 0.565 m with our method (relative improvements of 32.7% and 40.9%) demonstrates the superior capability of MBHNet in accurately recovering fine-grained building height information. Furthermore, given the inference times reported on the right-hand side of Tables I–IV, our method is also highly efficient, enabling rapid extraction of building heights and footprints. The qualitative results on the Test Set-I are presented in Fig. 11. Compared with other methods, the proposed MBHNet demonstrates more reliable

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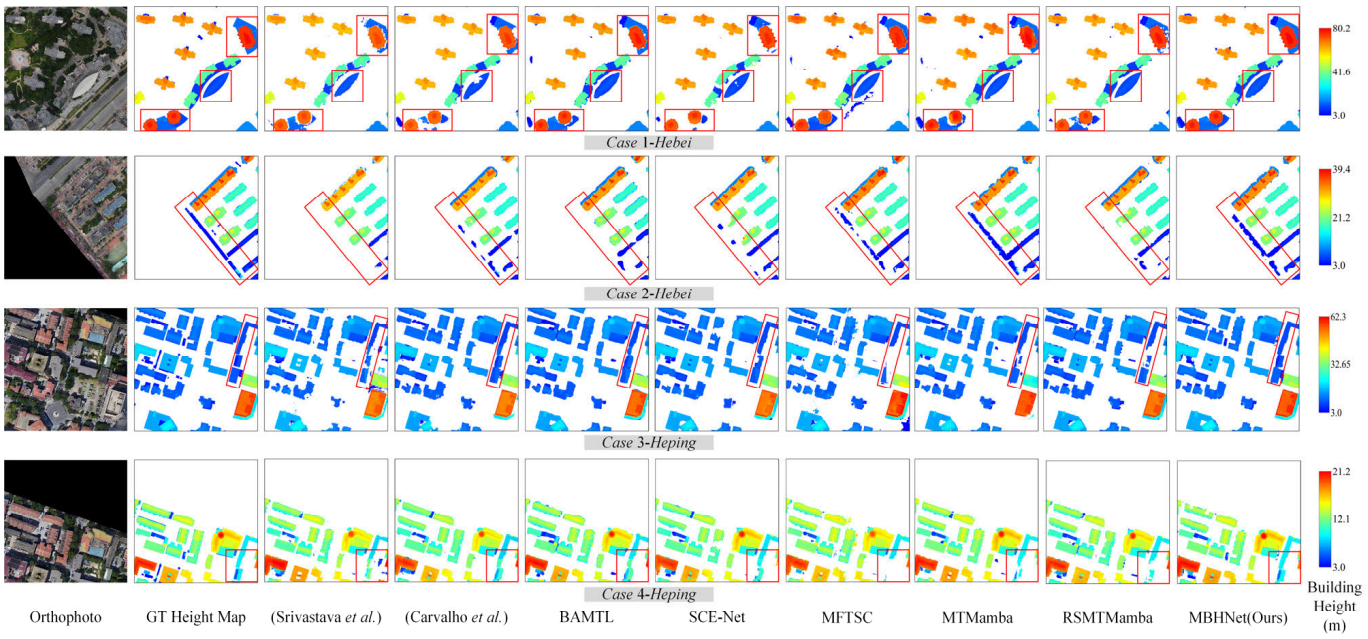


Fig. 12. Visualization of predicted building height maps generated by different comparison methods on the Test Set-II of Tianjin dataset.

TABLE III
QUANTITATIVE RESULTS OF DIFFERENT MTL METHODS ON THE TEST SET-II OF TIANJIN DATASET

Method	Height Estimation			Footprint Segmentation			Inference Time (seconds)
	MAE (m) ↓	RMSE (m) ↓	absRel ↓	IoU (%) ↑	F1 (%) ↑	OA (%) ↑	
Method by (*Srivastava et al.*, 2017)	0.901	1.570	0.051	82.8	90.6	96.1	4.35
Method by (*Carvalho et al.*, 2020)	0.932	1.220	0.054	84.3	91.5	96.4	**4.16**
BAMTL (Wang *et al.*, 2021)	0.538	1.179	0.037	84.7	91.7	96.5	5.26
SCE-Net (*Xing et al.*, 2022)	0.604	1.017	0.036	83.2	90.8	96.1	6.30
MFTSC (*Chen et al.*, 2023)	0.755	1.447	0.047	86.0	92.5	96.8	4.30
MTMamba (*Lin et al.*, 2024)	0.460	1.321	0.025	86.2	92.6	96.8	6.53
RSMTMamba (*Shen et al.*, 2025)	0.645	1.148	0.038	85.0	91.9	96.5	4.93
MBHNet (Ours)	**0.275**	**0.551**	**0.020**	**87.2**	**93.1**	**97.0**	4.82

height estimation (first two rows), more complete footprint segmentation (third row), and superior capability in preserving fine-grained structural details (fourth row).

3) Tianjin Dataset (Test-II): We utilized Test Set-II of Tianjin dataset to evaluate the generalization capability of different methods. Specifically, the models were trained on one region (Xiqing District) and tested on nearby but geographically isolated regions (Hebei District and Heping District). The quantitative results are reported in Table III, and corresponding visualizations of the predicted building height maps are presented in Fig. 12. The results show that most methods achieve comparable performance to that observed on Test Set-I. In general, slightly better height estimation performance, except for MTMamba [39], and slightly worse footprint segmentation performance, except for the methods of Srivastava et al. [14] and BAMTL [27], are observed when models trained from Xiqing are transferred to Hebei/Heping. These findings suggest that by labeling only a subset of the city for training, it is feasible to rapidly obtain reliable geometric and semantic information of the buildings for the entire city. Notably, our proposed MBHNet continues to demonstrate superior performance in both quantitative metrics and qualitative results, for example, achieving 40.2% MAE gain compared to the second-best method. As illustrated in Fig. 12, the prediction maps from MBHNet exhibit more accurate height estimation values and fewer missing footprint regions compared to those from other methods.

We also visualized the prediction results across the large images of Tianjin dataset (Test Set-I and Test Set-II). As illustrated in Fig. 13, although minor discrepancies remain, highlighted in the dashed box, the predicted building heights and footprint segmentations generally align well with the GT, demonstrating the strong performance of our MBHNet.

4) Potsdam Dataset: Table IV reports the quantitative results on Potsdam dataset. Consistent with the findings on the GeoNRW and Tianjin datasets, our proposed MBHNet achieves the best performance. For height estimation, MBHNet outperforms the second-best method (MTMamba) by 32.8% and 33.4% in terms of MAE and RMSE, respectively, and also achieves the highest footprint segmentation accuracy. The visual comparisons in Fig. 14 further illustrate that MBHNet produces building height and footprint estimates that are consistently closer to the GT than those from the other methods.

D. Ablation Analysis

MBHNet was designed in a modular manner. To evaluate the effectiveness of each component, we conduct ablation studies in a bottom-up fashion, starting from the encoding

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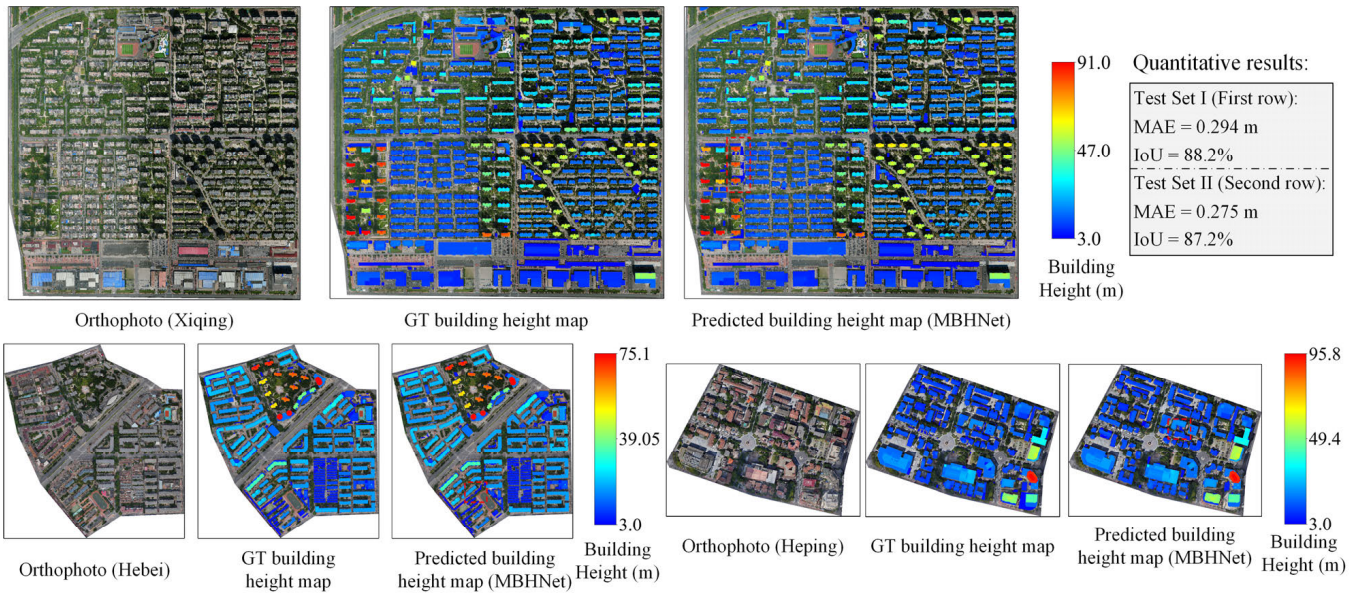


Fig. 13. Large-scale prediction results of MBHNet on Tianjin dataset (overlaid on orthophotos).

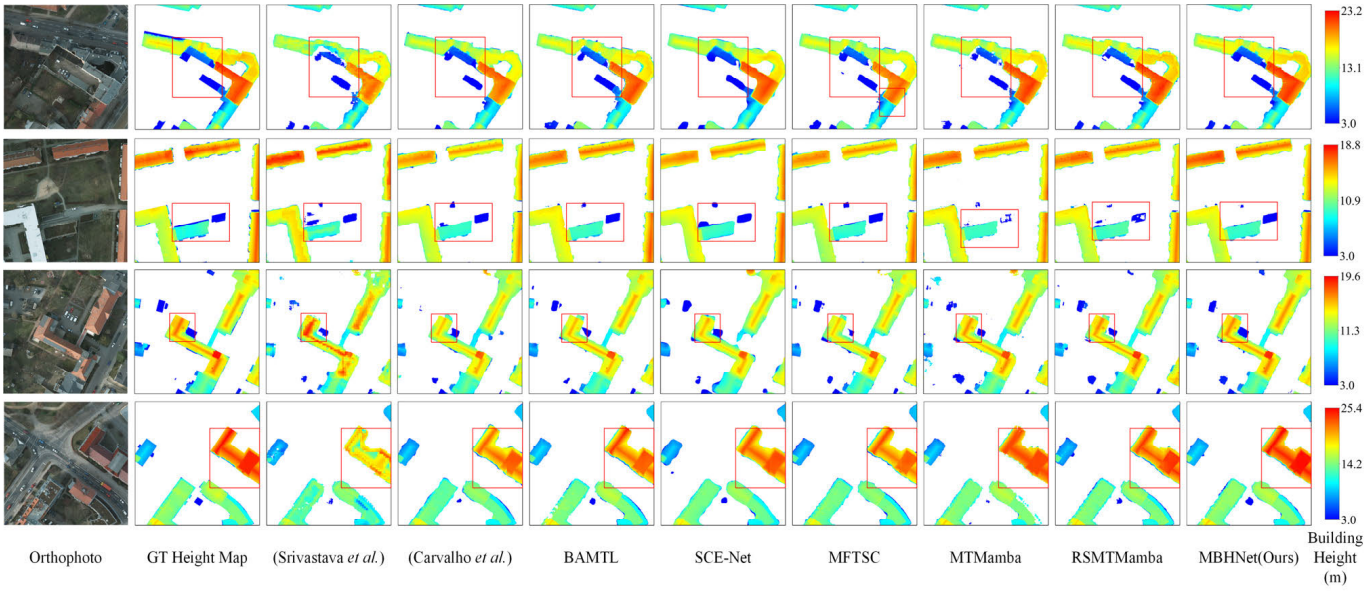


Fig. 14. Visualization of predicted building height maps generated by different comparison methods on Potsdam dataset.

TABLE IV
QUANTITATIVE RESULTS OF DIFFERENT MTL METHODS ON POTSDAM DATASET

Method	Height Estimation			Footprint Segmentation			Inference Time (seconds)
	MAE (m) ↓	RMSE (m) ↓	absRel ↓	IoU (%) ↑	F1 (%) ↑	OA (%) ↑	
Method by (*Srivastava et al.*, 2017)	1.471	2.416	0.140	82.4	90.4	95.4	9.01
Method by (*Carvalho et al.*, 2020)	1.156	2.336	0.101	88.1	93.7	97.0	**7.82**
BAMTL (Wang *et al.*, 2021)	0.848	2.099	0.074	89.4	94.4	97.3	10.18
SCE-Net (*Xing et al.*, 2022)	1.018	2.398	0.100	88.7	94.0	97.1	10.78
MFTSC (*Chen et al.*, 2023)	1.117	2.174	0.098	89.8	94.6	<u>97.4</u>	7.83
MTMamba (*Lin et al.*, 2024)	<u>0.692</u>	<u>1.540</u>	<u>0.069</u>	87.3	93.2	96.8	11.69
RSMTMamba (*Shen et al.*, 2025)	0.822	2.291	0.097	<u>90.4</u>	<u>95.0</u>	**97.6**	10.11
MBHNet (Ours)	**0.465**	**1.026**	0.046	**90.6**	**95.1**	**97.6**	9.71

stage and progressing to the decoding stage. For consistency and convenience, all ablation experiments in this section were conducted on Tianjin Test Set-I and II. This ablation study begins with a conventional MTL architecture comprising a shared encoder and two task-specific decoders for building height prediction and footprint segmentation.

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TABLE V
PERFORMANCE OF DIFFERENT FEATURE ENCODING STRUCTURES FOR BUILDING HEIGHT MAPPING (TIANJIN DATASET)

Dataset	Encoder	MAE (m) ↓	absRel ↓	IoU (%) ↑	F1 (%) ↑
Test-I	ResNet-50	1.320	0.077	83.2	90.8
	Swin Transformer	0.730	0.045	84.3	91.5
	VMamba	0.595	0.033	85.8	92.4
	LFS + LTL	0.621	0.039	83.9	91.3
	GMS + LTL	0.482	0.029	86.8	92.9
	LFS + GMS + LTL	**0.400**	**0.027**	**87.3**	**93.2**
Test-II	ResNet-50	1.209	0.063	85.7	92.3
	Swin Transformer	0.525	0.029	85.4	92.1
	VMamba	0.499	0.031	84.1	91.4
	LFS + LTL	0.581	0.031	84.6	91.7
	GMS + LTL	0.458	0.027	85.2	92.0
	LFS + GMS + LTL	**0.338**	**0.025**	**86.6**	**92.8**

TABLE VI
PERFORMANCE OF DIFFERENT BOTTLENECKS FOR BUILDING HEIGHT MAPPING (TIANJIN DATASET)

Dataset	Bottleneck	MAE (m) ↓	absRel ↓	IoU (%) ↑	F1 (%) ↑
Test-I	ConvBlock	0.400	0.027	87.3	93.2
	Channel attention (ECA)	0.389	0.026	87.2	93.2
	CSSM (Ours)	**0.351**	**0.025**	**87.4**	**93.3**
Test-II	ConvBlock	0.338	0.025	86.6	92.8
	Channel attention (ECA)	0.322	0.023	86.6	92.8
	CSSM (Ours)	**0.315**	**0.022**	**86.7**	**92.9**

TABLE VII
PERFORMANCE OF DIFFERENT FEATURE MINING RATIOS FOR BUILDING HEIGHT MAPPING (TIANJIN DATASET, TILE SIZE: 512 × 512 PIXELS)

Dataset	KTM	MAE (m) ↓	absRel ↓	IoU (%) ↑	F1 (%) ↑	Memory (Mb/Tile)
Test-I	K=0 (*None*)	0.351	0.025	87.4	93.3	439.27
	K=25%	0.309	0.025	88.3	**93.8**	511.71
	K=50%	**0.294**	**0.022**	88.2	93.7	511.82
	K=75%	0.333	0.025	**88.4**	**93.8**	511.94
	K=100%	0.328	0.025	88.1	93.7	512.05
Test-II	K=0 (*None*)	0.315	0.022	86.7	92.9	439.27
	K=25%	0.277	**0.020**	86.7	92.9	511.71
	K=50%	**0.275**	**0.020**	**87.2**	**93.1**	511.82
	K=75%	0.303	0.022	86.7	92.8	511.94
	K=100%	0.308	0.022	86.8	92.9	512.05

1) LS-SSM Encoder: We first evaluate the effectiveness of the proposed LS-SSM encoder. The results are reported in Table V, where LFS, GMS, and LTL denote the local focusing subblock, global modeling subblock, and linear transform layer, respectively.

First, compared to the most widely used ResNet-50 [61], which relies on a local receptive field, our encoder (LFS + GMS + LTL) achieves very significantly better height estimation accuracy (69.7% MAE gain in the Test-I and 72.0% in the Test-II) and improved footprint segmentation performance (4.1% IoU gain in the Test-I and 0.9% in the Test-II). Second, our encoder demonstrates much better performance than Swin Transformer [54] (45.2% MAE gain in the Test-I and 35.6% in the Test-II) and VMamba [20] (32.8% MAE gain in the Test-I and 32.3% in the Test-II), both of which also possess strong global modeling capabilities. This improvement can be attributed to the proposed IDE, which is the key distinction between GMS + LTL and VMamba. We see that even using only the combination of GMS and LTL, our encoder outperforms VMamba by 19.0%/8.2% in MAE for Test-I/II. The performance gap directly validates the design idea of our IDE: enhancing feature discrimination across different spatial regions prior to global modeling enables Mamba's input-dependent selection mechanism to function more effectively.

2) CSSM Bottleneck: To promote effective feature decoupling for both tasks, we further focus on the bottleneck within the MTL network. We take the commonly used bottleneck structure consisting of two convolutional blocks (each composed of a 3 × 3 convolutional layer, a batch normalization, and a ReLU activation) as the baseline, which serves as a bridge to transform shared encoder features into task-specific decoder features. As listed in Table VI, replacing this baseline bottleneck with the proposed CSSM leads to a notable reduction in MAE for height estimation, from 0.400 m/0.338 m to 0.351/0.315 m in Test Set-I and II, respectively, along with consistent improvements in footprint segmentation accuracy. In contrast, replacing the convolutional bottleneck with a popular channel attention-based bottleneck (efficient channel attention (ECA) [55]) yields only marginal or even negative performance gains. This outcome highlights the superior feature decoupling capability of CSSM, which effectively models interchannel dependencies without compressing the channel-wise information, alleviating a key limitation of typical channel attention mechanisms.

3) KTM Strategy: We introduce the KTM strategy to enable selective cross-task prompting during the feature decoding stage. As summarized in Table VII, the model performs differently when varying the proportion of mined features. Specifically, setting the hyperparameter K to 50% yields the best performance for building height prediction on both Test-I and Test-II, as well as the best footprint segmentation performance on Test-II, across the range from no sharing (0%) to full sharing (100%), evaluated at 25% intervals. Increasing K to 75% leads to the highest footprint segmentation accuracy in Test-I. Notably, when K = 100%, which corresponds to full feature sharing and is widely used in recent MTL frameworks [9], [27], [28], [39], the model fails to achieve optimal performance for either task. This observation confirms that not all features from one task are beneficial to the other; excessive feature sharing can introduce redundancy and even lead to undesirable feature recoupling between tasks. Considering the tradeoff between the two tasks, we ultimately adopt K = 50% as a reasonable choice in the proposed method. Note that using finer-grained intervals to determine K could further improve results.

4) Hyperparameters in the Loss Function: We introduce two hyperparameters into the loss function to balance auxiliary supervision for cross-task feature transfer with final output supervision. We conduct an ablation study here to determine a reasonable setting, where the loss weight α for the final output supervision is fixed to 1, and the auxiliary supervision weight β is varied. As shown in Table VIII, the performance degrades without auxiliary supervision (β = 0). Meanwhile, we observe that the overall performance changes only slightly as β varies. A small auxiliary weight of β = 0.05 yields

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TABLE VIII

PERFORMANCE OF DIFFERENT LOSS FUNCTION SETTINGS FOR BUILDING HEIGHT MAPPING (TIANJIN DATASET, $\alpha = 1$, $K = 50\%$)

Dataset	Parameter	MAE (m) ↓	absRel ↓	IoU (%) ↑	F1 (%) ↑
Test-I	β=0	0.335	0.023	87.2	93.2
	β=0.05	**0.294**	**0.022**	**88.2**	**93.7**
	β=0.10	0.301	0.023	87.8	93.5
	β=0.20	0.300	0.023	87.7	93.5
Test-II	β=0	0.313	0.020	87.0	**93.1**
	β=0.05	0.275	0.020	**87.2**	**93.1**
	β=0.10	**0.273**	**0.019**	86.8	92.9
	β=0.20	0.278	**0.019**	87.0	**93.1**

TABLE IX

COMPUTATIONAL COSTS AND MODEL PARAMETERS OF DIFFERENT MTL METHODS

Method	GFLOPs	Params (MB)
Method by (*Srivastava et al.*, 2017)	422.0	**4.00**
Method by (*Carvalho et al.*, 2020)	252.7	24.7
BAMTL (*Wang et al.*, 2021)	343.6	48.5
SCE-Net (*Xing et al.*, 2022)	386.0	166.2
MFTSC (*Chen et al.*, 2023)	75.8	34.9
MTMamba (*Lin et al.*, 2024)	**52.3**	42.8
RSMTMamba (*Shen et al.*, 2025)	124.4	30.9
MBHNet (Ours)	121.5	48.0

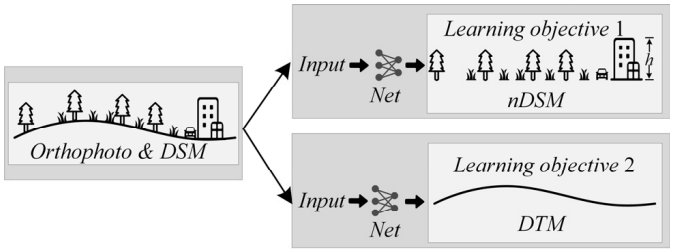


Fig. 15. Two types of learning objectives for MTL models in recovering building heights from orthophotos and DSMs, in the bottom strategy, building height h = DSM − DTM.

excellent performance, whereas further increasing this weight leads to a slight performance decline in footprint segmentation. Therefore, we set $\alpha = 1$ and $\beta = 0.05$ in our method.

5) Model Complexity: Table IX summarizes the computational costs and model sizes of all compared methods. MBHNet requires 121.5 GFLOPs and 48.0 MB of parameters, representing a moderate level among the methods. Although MTMamba [39] has the lowest GFLOPs (52.3) and competitive parameter size (42.8 MB), it suffers from significantly longer inference times, as shown in the last columns of Tables I–III. Compared with parameter-heavy models like SCE-Net [28] (166.2 MB) and computation-intensive models like the method of Srivastava et al. [14] (422.0 GFLOPs), MBHNet achieves a good balance between model complexity, computational cost, inference efficiency, and key accuracy, further demonstrating the advancement of our design. Inferring on the approximately 380-km^2 GeoNRW test set, MBHNet takes less than 1 min, as listed in Table I.

E. Discussion

1) Different Learning Objectives: As illustrated in Fig. 15, building height recovery within MTL frameworks can be formulated using two different learning objectives: directly predicting the nDSM or indirectly learning the DTM and subtracting it from the DSM, both restricted to building regions. In this section, we compare the performance of the two learning objectives using the proposed MBHNet. To further analyze the DTM-learning model, we additionally compared it with a simple empirical baseline, where ground elevation is estimated from the elevation histogram by selecting the value corresponding to a small cumulative frequency (5%, following [8]). The quantitative results are summarized in Table X.

TABLE X

PERFORMANCE COMPARISON WITH DIFFERENT LEARNING OBJECTIVES FOR BUILDING HEIGHT MAPPING

Dataset	Learning objective	MAE(m) ↓	absRel ↓	IoU (%) ↑	F1 (%) ↑
Tianjin Test-I	nDSM	0.294	**0.022**	**88.2**	**93.7**
	DTM	**0.258**	0.024	87.1	93.1
	Simple baseline	0.629	0.048	87.1	93.1
Tianjin Test-II	nDSM	**0.275**	**0.020**	**87.2**	**93.1**
	DTM	0.985	0.075	83.4	91.0
	Simple baseline	0.521	0.032	83.4	91.0
GeoNRW	nDSM	**0.627**	**0.065**	**79.3**	**88.4**
	DTM	0.949	0.103	78.4	87.9
	Simple baseline	4.296	0.551	78.4	87.9

TABLE XI

PERFORMANCE COMPARISON OF DIFFERENT MTL METHODS FOR MONOCULAR BUILDING HEIGHT MAPPING

Method	MAE (m) ↓	absRel ↓	IoU (%) ↑	F1 (%) ↑
MTMamba (2024)	2.007	0.210	67.8	80.8
RSMTMamba (2025)	1.947	0.208	70.8	82.9
MBHNet (Ours)	**1.866**	**0.198**	**74.4**	**85.3**
MBHNet (DSM+orthophotos)	**0.640**	**0.064**	**79.5**	**88.6**

On Tianjin Test Set-I, the DTM-learning model achieves comparable accuracy to the nDSM-learning model for height estimation, though it yields slightly lower performance in footprint segmentation. However, on the Test Set-II, the DTM-learning model shows notably poorer generalization, resulting in significantly worse outcomes for both height prediction and footprint segmentation. The GeoNRW dataset, unlike the relatively flat terrain of Tianjin, features more diverse topography with greater elevation variation. On this dataset, the DTM-learning model again significantly underperforms the nDSM-learning model in terms of height estimation, and also performs slightly worse in footprint segmentation.

Compared with the simple baseline, the DTM-learning model produces more accurate heights on Tianjin Test-I and the GeoNRW dataset, demonstrating that it can learn meaningful DTM representations under similar domain conditions. On Tianjin Test-II, however, the DTM-learning model performs worse than the simple baseline, suggesting that the learned DTM representations fail to generalize under cross-region domain shifts.

As shown in Fig. 16, we visualized error maps of nDSM-based and DTM-based pipelines on Tianjin Test-II. The DTM-learning model exhibits significantly larger absolute errors and distinct region-dependent error structures, with the middle road acting as a clear separation line. This indicates that the DTM objective is highly sensitive to regional terrain

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TABLE XII
QUANTITATIVE RESULTS OF DIFFERENT MTL METHODS ON THE TEST SET OF THE URBAN 3-D CHALLENGE DATASET

Method	Height Estimation			Footprint Segmentation			Inference Time (seconds)
	MAE (m) ↓	RMSE (m) ↓	absRel ↓	IoU (%) ↑	F1 (%) ↑	OA (%) ↑	
MTMamba (*Lin et al.*, 2024)	0.734	1.643	0.010	64.5	78.4	95.1	63.27
RSMTMamba (*Shen et al.*, 2025)	0.723	1.533	0.101	68.1	81.0	95.6	44.92
MBHNet (Ours)	**0.644**	**1.371**	**0.091**	**71.3**	**83.3**	**96.0**	41.53



Fig. 16. (a) Reference image from Tianjin Test-II (Hebei District). (b) Prediction error map from the DTM-learning model. (c) Prediction error map from the nDSM-learning model.

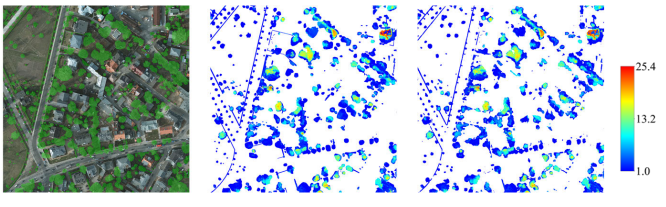


Fig. 17. Tree-height mapping results on Potsdam dataset. (a) Input image where tree-covered regions are highlighted with a light green overlay, (b) GT height map, and (c) height map predicted by MBHNet.

characteristics, leading to strong location-dependent biases. In contrast, the nDSM-learning model shows no such spatially clustered errors and maintains stable performance across regions.

2) Monocular Building Height Estimation: While our proposed method utilizes both DSMs and orthophotos for building height recovery, many comparison methods were originally designed for monocular height estimation. To further verify the effectiveness of the proposed method, we conducted experiments when only orthophotos are available, leveraging the large-scale GeoNRW dataset to compensate for the increased learning difficulty. As reported in Table XI, omitting the DSM leads to a decrease in all metrics. Nevertheless, our method still achieves slightly better height estimation performance than two recent monocular methods, MTMamba [39] and RSMTMamba [9]. Moreover, our method delivers significantly better footprint segmentation results. This finding demonstrates that the advantages of our designed framework still exist in monocular settings.

3) Cross-City Building Height Inference: In the previous experiments, including those conducted on the GeoNRW, Tianjin, and Potsdam datasets, height inference was evaluated only within individual cities (the GeoNRW dataset contains multiple mixed cities). To further assess cross-city generalization capability, we conducted a dedicated cross-city height inference experiment.

The Urban 3-D Challenge dataset [63] was used for this purpose. It provides orthorectified RGB images and DSMs/DTMs with a GSD of 0.5 m, covering approximately 278 km^2 and containing 125 772 buildings across three U.S. cities: Tampa, Florida; Jacksonville, Florida; and Richmond, Virginia. The training/validation set consists of 174/62 tiles of size 2048 × 2048 pixels, sampled from Tampa and Jacksonville, while the test set includes 42 tiles exclusively from Richmond, thereby enabling strict cross-city evaluation.

MTMamba [39] and RSMTMamba [9] were selected as comparison baselines due to their competitive performance in the previous experiments. The quantitative results are presented in Table XII. In the cross-city height inference setting, our method significantly outperforms the two baselines, achieving substantially lower height estimation errors and higher footprint segmentation accuracy, which further demonstrates its strong generalization capability.

4) Limitations and Potential Applications: In urban datasets, building height is typically computed by subtracting the DTM from the DSM within each building footprint, as done in the ISPRS Vaihingen dataset processed by Gerke [62] and in this work. Under this convention, when tree canopy overlaps building roofs, the canopy returns are treated as part of the elevated building surface, resulting in overestimated building heights.

The proposed method learns to estimate the height from the ground to the object's uppermost point within its footprint. Therefore, the proposed method is also adapted for forest canopy-height mapping, where tree crowns are the objects of interest, provided that portions of bare ground are visible to serve as a height reference. To demonstrate this potential application, we experimented on the tree category of the ISPRS Potsdam dataset using the proposed method. MBHNet achieved an MAE of 0.547 m and an IoU of 71.2%, and the resulting tree-height map is shown in Fig. 17. The prediction shows good consistency with the GT map, indicating promising generalization.

V. CONCLUSION

This article proposes a novel MTL method called MBHNet for building height estimation from orthophotos and DSMs, coupling height prediction and footprint segmentation. MBHNet integrates a LS-SSM for data-customized feature encoding, a channel-dimension SSM for sharing feature decoupling, and a KTM for selective cross-task prompting. These designs enable the production of fine-grained building height and footprint maps. Both qualitative and quantitative results confirm that MBHNet significantly outperforms other recent

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methods on the GeoNRW, Tianjin, and Potsdam datasets. Ablation studies validated the effectiveness of each component in MBHNet. These explorations collectively contribute to advancing building height estimation research.

REFERENCES

[1] D. Frantz et al., "Unveiling patterns in human dominated landscapes through mapping the mass of U.S. built structures," *Nature Commun.*, vol. 14, no. 1, p. 8014, Dec. 2023.

[2] W.-B. Wu et al., "A first Chinese building height estimate at 10 m resolution (CNBH-10 m) using multi-source Earth observations and machine learning," *Remote Sens. Environ.*, vol. 291, Jun. 2023, Art. no. 113578.

[3] D. Yu, S. Ji, S. Wei, and K. Khoshelham, "3-D building instance extraction from high-resolution remote sensing images and DSM with an end-to-end deep neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4406019.

[4] Y. Yan and B. Huang, "Estimation of building height using a single street view image via deep neural networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 192, pp. 83–98, Oct. 2022.

[5] T. Leichtle, T. Lakes, X. X. Zhu, and H. Taubenböck, "Has Dongying developed to a ghost city?—Evidence from multi-temporal population estimation based on VHR remote sensing and census counts," *Comput., Environ. Urban Syst.*, vol. 78, Nov. 2019, Art. no. 101372.

[6] W.-B. Wu, Z.-W. Yu, J. Ma, and B. Zhao, "Quantifying the influence of 2D and 3D urban morphology on the thermal environment across climatic zones," *Landscape Urban Planning*, vol. 226, Oct. 2022, Art. no. 104499.

[7] Y. Zhang, H. Zhao, and Y. Long, "CMAB: A multi-attribute building dataset of China," *Scientific Data*, vol. 12, no. 1, p. 430, Mar. 2025.

[8] C. Zeng, J. Wang, W. Zhan, P. Shi, and A. Gambles, "An elevation difference model for building height extraction from stereo-image-derived DSMs," *Int. J. Remote Sens.*, vol. 35, no. 22, pp. 7614–7630, Nov. 2014.

[9] Y. Shen, L. Xiao, J. Chen, Q. Du, and Q. Ye, "Learning cross-task features with mamba for remote sensing image multitask prediction," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5612116.

[10] S. Wang, B. Cai, D. Hou, Q. Ding, J. Wang, and Z. Shao, "MF-BHNet: A hybrid multimodal fusion network for building height estimation using Sentinel-1 and Sentinel-2 imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4512419.

[11] M. Barranquero, A. Olmedo, J. Gómez, A. Tayebi, C. J. Hellín, and F. Saez de Adana, "Automatic 3D building reconstruction from OpenStreetMap and LiDAR using convolutional neural networks," *Sensors*, vol. 23, no. 5, p. 2444, Feb. 2023.

[12] D. Yu, S. Ji, J. Liu, and S. Wei, "Automatic 3D building reconstruction from multi-view aerial images with deep learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 171, pp. 155–170, Jan. 2021.

[13] Y. Yan, F. Gao, S. Deng, and N. Su, "A hierarchical building segmentation in digital surface models for 3D reconstruction," *Sensors*, vol. 17, no. 2, p. 222, Jan. 2017.

[14] S. Srivastava, M. Volpi, and D. Tuia, "Joint height estimation and semantic labeling of monocular aerial images with CNNS," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 5173–5176.

[15] M. Carvalho, B. Le Saux, P. Trouvé-Peloux, F. Champagnat, and A. Almansa, "Multitask learning of height and semantics from aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 8, pp. 1391–1395, Aug. 2020.

[16] F. Alidoost, H. Arefi, and F. Tombari, "2D image-to-3D model: Knowledge-based 3D building reconstruction (3DBR) using single aerial images and convolutional neural networks (CNNs)," *Remote Sens.*, vol. 11, no. 19, p. 2219, Sep. 2019.

[17] D. Wang, Q. Jiang, and J. Liu, "Deep-learning-based automated building information modeling reconstruction using orthophotos with digital surface models," *Buildings*, vol. 14, no. 3, p. 808, Mar. 2024.

[18] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," in *Proc. Conf. Lang. Model.*, Milano, Italy, Aug. 2023, pp. 1–12.

[19] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," in *Proc. Int. Conf. Learn. Rep. (ICLR)*, Jan. 2021, pp. 1–21.

[20] Y. Liu et al., "VMamba: Visual state space model," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 37, 2024, pp. 103031–103063.

[21] Y. A. Mousa, P. Helmholz, and D. Belton, "New dtm extraction approach from airborne images derived DSM," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 1, pp. 75–82, May 2017.

[22] S. Zhao, H. Chen, X. Zhang, P. Xiao, L. Bai, and W. Ouyang, "RS-Mamba for large remote sensing image dense prediction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5633314.

[23] X. Ma, X. Zhang, and M.-O. Pun, "RS^{3}Mamba: Visual state space model for remote sensing image semantic segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[24] A. Hassani, S. Walton, J. Li, S. Li, and H. Shi, "Neighborhood attention transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 6185–6194.

[25] Z. Gao et al., "Joint learning of semantic segmentation and height estimation for remote sensing image leveraging contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5614015.

[26] Y. Chen, Q. Yan, and W. Huang, "MFTSC: A semantically constrained method for urban building height estimation using multiple source images," *Remote Sens.*, vol. 15, no. 23, p. 5552, Nov. 2023.

[27] Y. Wang, W. Ding, R. Zhang, and H. Li, "Boundary-aware multitask learning for remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 951–963, 2021.

[28] S. Xing, Q. Dong, and Z. Hu, "SCE-Net: Self-and cross-enhancement network for single-view height estimation and semantic segmentation," *Remote Sens.*, vol. 14, no. 9, p. 2252, May 2022.

[29] M. Wang, Z. Yan, Y. Feng, W. Diao, and X. Sun, "Multi-task learning of semantic segmentation and height estimation for multi-modal remote sensing images," *J. Geod. Geoinf. Sci.*, vol. 6, no. 4, pp. 27–39, 2023.

[30] H. Ye and D. Xu, "InvPT: Inverted pyramid multi-task transformer for dense scene understanding," in *Proc. Eur. Conf. Comput. Vis.*, Jan. 2022, pp. 1–16.

[31] N. Carion, F. Massa, and G. Synnaeve, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, vol. 2020, pp. 213–229.

[32] B. Cai, Z. Shao, X. Huang, X. Zhou, and S. Fang, "Deep learning-based building height mapping using Sentinel-1 and Sentinel-2 data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 122, Aug. 2023, Art. no. 103399.

[33] C. Ji and H. Tang, "Number of building stories estimation from monocular satellite image using a modified mask R-CNN," *Remote Sens.*, vol. 12, no. 22, p. 3833, Nov. 2020.

[34] J. Wu, M. Yuan, T. Wang, X. Jia, and D.-M. Yan, "HeightFormer: Single-imagery height estimation transformer with bilateral feature pyramid fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[35] S. Chen, Y. Shi, Z. Xiong, and X. X. Zhu, "HTC-DC Net: Monocular height estimation from single remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5623018.

[36] S. Chen, L. Mou, Q. Li, Y. Sun, and X. X. Zhu, "Mask-height R-CNN: An end-to-end network for 3D building reconstruction from monocular remote sensing imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. IGARSS*, Jul. 2021, pp. 1202–1205.

[37] S. Baek, S. Lee, H. Jo, H. Choi, and D. Min, "TADFormer: Task-adaptive dynamic transformer for efficient multi-task learning," 2025, *arXiv:2501.04293*.

[38] H. Huang et al., "Going beyond multi-task dense prediction with synergy embedding models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 28181–28190.

[39] B. Lin, W. Jiang, P. Chen, Y. Zhang, S. Liu, and Y.-C. Chen, "MTMamba: Enhancing multi-task dense scene understanding by Mamba-based decoders," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024, pp. 314–330.

[40] Y. Tang, S. Feng, C. Zhao, Y. Chen, Z. Lv, and W. Sun, "A semantic change detection network based on boundary detection and task interaction for high-resolution remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 9, pp. 17184–17198, Sep. 2025.

[41] J. Wang, T. Gangavarapu, J. N. Yan, and A. M. Rush, "MambaByte: Token-free selective state space model," 2024, *arXiv:2401.13660*.

[42] Z. Xing, T. Ye, Y. Yang, G. Liu, and L. Zhu, "SegMamba: Long-range sequential modeling mamba for 3D medical image segmentation," 2024, *arXiv:2401.13560*.

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[43] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision Mamba: Efficient visual representation learning with bidirectional state space model," in *Proc. Int. Conf. Mach. Learn. (ICML)*, Jan. 2024, pp. 1–14.

[44] X. He et al., "Pan-mamba: Effective pan-sharpening with state space model," *Inf. Fusion*, vol. 115, Mar. 2025, Art. no. 102779.

[45] K. Chen, B. Chen, C. Liu, W. Li, Z. Zou, and Z. Shi, "RSMamba: Remote sensing image classification with state space model," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[46] Q. Zhu, Y. Fang, Y. Cai, C. Chen, and L. Fan, "Rethinking scanning strategies with vision mamba in semantic segmentation of remote sensing imagery: An experimental study," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 18223–18234, 2024.

[47] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "3DSS-Mamba: 3D-spectral–spatial Mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534216.

[48] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409720.

[49] E. Zhu, Z. Chen, D. Wang, H. Shi, X. Liu, and L. Wang, "UNetMamba: An efficient UNet-like mamba for semantic segmentation of high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[50] C. Zhu, S. Deng, X. Song, Y. Li, and Q. Wang, "Mamba collaborative implicit neural representation for hyperspectral and multispectral remote sensing image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504915.

[51] Y. Li et al., "MambaHSI: Spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216.

[52] G. Wang et al., "S^2Mamba: A spatial–spectral state space model for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5511413.

[53] J. Li, Y. Wen, and L. He, "SCConv: Spatial and channel reconstruction convolution for feature redundancy," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 6153–6162.

[54] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[55] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient channel attention for deep convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11531–11539.

[56] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.

[57] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 565–571.

[58] G. Baier, A. Deschemps, M. Schmitt, and N. Yokoya, "Synthesizing optical and SAR imagery from land cover maps and auxiliary raster data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4701312.

[59] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2012, pp. 1097–1105.

[60] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2261–2269.

[61] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[62] M. Gerke, "Use of the stair vision library within the ISPRS 2D semantic labeling benchmark (Vaihingen)," Dept. Earth Observ. Sci., Univ. Twente, Enschede, The Netherlands, Tech. Rep., 2014, doi: 10.13140/2.1.5015.9683.

[63] H. R. Goldberg et al., "Urban 3D challenge: Building footprint detection using orthorectified imagery and digital surface models from commercial satellites," *Proc. SPIE*, vol. 10645, pp. 12–31, Apr. 2018.

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